

Utilization of semantic web technologies to improve BIM-LCA applications

Soroush Sobhkhiz, Hossein Taghaddos^{*}, Mojtaba Rezvani, Amir Mohammad Ramezaniapour

School of Civil Engineering, College of Engineering, University of Tehran, Tehran, Iran

ARTICLE INFO

Keywords:

Semantic Web
SPARQL
BIM
LCA
SQL
GHG emissions
Cradle to site analysis

ABSTRACT

Currently, designers require to use numerous incomplete data sources and make multiple assumptions to conduct an LCA. We argue that traditional methods are incapable of addressing LCA data requirements, and for such a data and meta-data intensive task, the semantic web is the only logical choice of technology. This paper presents a review of the semantic web and its applications in the construction industry, along with an overview of the current LCA tools. LCA challenges are thoroughly discussed, and a framework using the semantic web is developed to address the data management issues of BIM-LCA applications. For comparison, the implementation is also done using an LCA tool (Tally), and a method using Relational Database Management Systems (RDBMS). Results show that the proposed method is superior to RDBMS methods in terms of capturing semantics and can improve LCA tools by providing reliable data in the early design stages.

1. Introduction

The role of the construction industry in harming the environment is very well-known. “Consuming 32% of global resources”, “the main waste producer in the world”, “responsible for over 40% of global GHG emissions”, and many other facts are all highlighting the need for a more sustainable industry [1]. Although the question of “how to make the construction industry a sustainable one” may not be a state-of-the-art issue, it certainly demands state-of-the-art solutions.

There are currently several tools and methods to help incorporate sustainable development into the construction industry, among which Life Cycle Assessment (LCA) has a growing recognition and importance in the scientific community. LCA deals with assessing the sustainability of a building over its life cycle. The life cycle refers to all stages, from raw material extraction to building demolition and material reuse/recycle. Therefore, LCA is significantly data-intensive and requires access to a broad spectrum of data from heterogeneous sources. LCA is considerably time-consuming and costly to implement, even in its very first stages. As a result, numerous studies over the past decade have focused on the improvement and simplification of LCA [1]. The integration of Building Information Modeling (BIM) with LCA practices is perhaps one of the most promising ways of optimizing LCA applications.

BIM can be used in various ways with various purposes for LCA applications. For example, a building’s digitalized model allows assessing its different aspects during its life cycle, including building orientation,

building massing, daylighting, water harvesting, energy modeling, using renewable energy, and employing sustainable materials. In general, BIM-LCA studies primarily focus on:

- 1- Developing a tool to quantify building materials in Life Cycle Inventory (LCI) stages,
- 2- Developing a tool to integrate environmental information with purposes such as energy evaluation,
- 3- Automating calculation processes by combining different data sources and analysis tools [2].

Assessing these studies can help explain the primary motivation behind the use of BIM in LCA practices; data management. By nature, BIM is an object-oriented database, and its capabilities in generating, storing, managing, and exchanging building information, provide valuable opportunities for LCA practices [3]. However, integrating BIM with a knowledge management system might be how we can take these studies to the next level.

1.1. Knowledge management and the semantic web

According to Liebowitz [4], knowledge is obtained by analyzing and organizing information. Knowledge is the result of understanding information, which is then used for problem-solving and decision-making. Over the past few decades, the advancement of the construction industry

^{*} Corresponding author.

E-mail addresses: sorouhsobhkhiz@ut.ac.ir (S. Sobhkhiz), HTaghaddos@ut.ac.ir (H. Taghaddos), m.rezvani723@ut.ac.ir (M. Rezvani), ramezani@ut.ac.ir (A.M. Ramezaniapour).

<https://doi.org/10.1016/j.autcon.2021.103842>

Received 20 October 2020; Received in revised form 1 July 2021; Accepted 21 July 2021

Available online 31 July 2021

0926-5805/© 2021 Elsevier B.V. All rights reserved.

has led to the creation of a large amount of Knowledge [5]. The need for the efficient management of this knowledge has motivated many studies to promote “Knowledge Management (KM)”. KM deals with the creation, acquisition, organization, usage, and evolution of Knowledge. Nowadays, KM tools are utilized in daily business processes and are integral parts of organizations. Effective KM is essential for the sustainable success of any enterprise. It can be argued that KM can be significantly effective in alleviating the construction industry’s sustainability standards. Specifically, there are unexplored potentials in applying KM techniques/technologies in LCA studies [6].

In the past two decades, there has been a shift towards AI-based techniques to redirect KM from humans to computers. The increasing interactions between humans and machines and the growing complexity of software application domains have all contributed to the recognition of this transformation [5]. To this end, efforts were made to develop techniques to represent Knowledge in ways that can be managed by machines. Such an approach resulted in the automatic generation of machine-readable and added-value data (i.e., Knowledge) and was the primary motivation behind the advent of the “Semantic Web”.

The Semantic Web is an extension of the current web to promote common data formats so that data can be read directly by computers. In the semantic web, information is accessed based on meaning rather than syntax, allowing computers to read the data and obtain Knowledge. The main idea is that using the semantic web will enable machines to analyze data independently and interact with other machines and even humans [7].

From a practical viewpoint, the semantic web has been fully embraced in many industries. However, its applications in the construction industry are scattered and largely unexplored [8]. This can be attributed to the lack of motivation from the industrial practitioners to make investments for broad applications of semantic web technologies. Perhaps, similar to the beginning ages of BIM, there is yet some time needed for the construction industry to realize the valuable potentials provided by the semantic web technologies to make substantial investments. Therefore, the scientific community’s role is to explore these potentials, showcase their practical applications. Since the advent of the semantic web, numerous research projects have been conducted to explore ways to utilize these technologies in the construction processes. These research studies vary from developing ontologies in the earlier years (i.e., the 2000s) to more advanced applications in recent years. An analysis of the literature will be provided in the literature review section. Despite the significant progress in recent years, there are still knowledge gaps in the current body of knowledge regarding the practical applications of semantic web technologies in sustainable development. More specifically, the potentials of the semantic web in addressing BIM-LCA limitations are still unexplored.

1.2. The semantic web for BIM-LCA

The integration of BIM and LCA is essential because it allows taking the environmental impact aspect into account for complex decision-making throughout a building’s life cycle. So far, researchers have mainly applied BIM to extract quantities to establish LCIs for LCA [6]. Nevertheless, most early design stage BIM-LCA studies have encountered the same challenge as the analysis conducted without BIM: information required to perform a complete LCA does not exist in BIM models, particularly in their early stages. Thus, an early-stage BIM-based LCA calls for significant manual input, making it complex, time-consuming, and even error-prone. This is where the semantic web can play a critical role.

The complexity of assessing a building’s life cycle environmental impacts originates from the construction industry’s fragmented nature and inherent diversity. Construction projects involve multiple stakeholders and deal with many products over various life cycle stages interacting dynamically in both time and space. As a result, planners must consider data (e.g., material properties and manufacturers’

information) from distributed information sources when conducting LCA in the early design stages. By providing intelligent solutions to manage the extensive amount of required information mainly by machines rather than humans, the semantic web can resolve these challenges and thus facilitate the path towards a more sustainable industry. Here, the long-term vision is to encourage the development of semantic-driven frameworks utilizing smart agents so that machines could conduct the analyses on their own. Of course, such solutions will not emerge for the foreseeable future, and there is a need for more studies paving the way for future developments. To this end, this research aims to propose a framework to leverage semantic web technologies for BIM-integrated LCA.

1.3. What is and what is not?

This research aims to establish a framework in which semantic web technologies are used as the primary knowledge management solution to facilitate sustainable construction. The framework’s objective is to illustrate how the semantic web can be used as a knowledge management tool to link different sources of data (e.g., BIM, databases, best-practices). It is crucial to understand that this research is not an attempt to propose new LCA methods or develop new ontologies. Most importantly, the proposed solution is not to replace current well-developed LCA tools but to address the data challenges associated with them and work with them as a complementary solution. The objective is merely to explain why the construction industry requires to utilize semantic knowledge bases at the core of its projects when it comes to green construction and then propose a way to do so.

As stated before, LCA is a vast concept encompassing all stages of a building life cycle. It is not feasible to address all of these complex processes in a single research study. Therefore, since the purpose is to showcase how the semantic web can be utilized in a BIM-LCA context, the LCA scope in this study is limited to the cradle-to-site stages. In other words, the proposed framework aids designers in analyzing the environmental impacts of the materials, from material extraction/production to transportation to the construction site.

2. Literature review

A brief review of the semantic web, its concepts, its applications, and trends in the construction industry is provided in this section. The state-of-the-art research is explained, and the gap and necessity to apply the semantic web technologies in sustainable construction are justified. In addition, an overview of the current LCA tools, their applications, and their limitations is presented. A case study using one of the LCA tools is also provided to showcase current advancements and limitations in LCA. To enrich the context, two LCA experts are also interviewed to bring an industrial perspective to the research.

2.1. An overview of the semantic web technology

In the semantic web, ontologies are used to provide a reference for computers to understand the meaning of things (i.e., concepts). According to Gruber [9], an ontology is a formal specification of a shared conceptualization. In other words, an ontology defines the concepts and the relationships between them in a hierarchical structure. There are numerous studies on how to develop an ontology, and multiple methods have been proposed [10], which is out of the scope of this research. The language used for developing ontologies is called Web Ontology Language, abbreviated as OWL. There are currently a number of ontology editors that can be used for writing ontologies in OWL, among which Protégé is one of the most reputable ones. In addition to ontologies, the semantic web uses Resource Description Framework (RDF) for data modeling. In RDF, data is modeled using triples of object, predicate, and subject. For instance, the following structure can be used to describe the density of a material such as concrete in RDF;

Subject: Concrete Predicate: hasDensity Object: 2000 kg/m³

With the above form, the entities do not have any semantics associated with them. In other words, a computer cannot understand what “hasDensity” means. Therefore, the entities should be linked to an ontology for the computer to understand the data’s meaning. The semantic web uses Unique Resource Identifiers (URIs) to link entities to their relevant concepts (i.e., relationships).

Now, to access the information and analyze/ update them, a new query language is required. The semantic web proposes using a popular RDF query language, called SPARQL (SPARQL Protocol and RDF Query language), to retrieve and manipulate RDF data. In short, the semantic web is a knowledge management approach that aims to make data machine-readable by storing data in RDF format and linking concepts to ontologies using URIs. The information is then accessible using the SPARQL query language. Fig. 1 shows the overall structure of how the semantic web represents information.

The Semantic web has two advantages [11]. The first one is that by providing a unique way to represent information, computers can understand the information and access the knowledge within it. Second, RDF data is easy to integrate and combine. Because data is eventually modeled in a graph-based structure, integrating two different data sources is simply finding the common nodes between them and then linking them together.

2.2. Why not the traditional methods?

It is a known fact that data and information are different. Information is interpreted data, organized in structured models (i.e., information models), in which data is accompanied by metadata about itself. Currently, the vast majority of enterprise applications manage data using Relational Database Management Systems (RDBMS), Unified Model Language (UML), and/or XML. However, these technologies were not primarily designed for the challenges of today’s world. For instance, RDBMS was built to store and retrieve data records, UML for a unified way of software structure visual modeling, and XML for structuring documents and messages. These technologies are not purposefully designed to specify data relationships across separate data locations or specify dynamic structural logic and rules [12]. Although we can use

these technologies for such purposes, the result will be (as currently is) one-off, non-standard solutions with little interoperability and portability. On the other hand, the semantic web is purposefully designed to deal with large-scale data problems. Specifications such as RDF, OWL, and SPARQL define consistent information models allowing declarative interaction with data [12].

2.3. Semantic web in the construction industry

In the literature, multiple studies have reviewed the trends and applications of semantic web technologies in the Architecture, Engineering, and Construction (AEC) industry. Going through these research studies is, therefore, a redundant task. However, the purpose of this section is to build an introductory foundation for the next two sections in which the research gap and successively the research question is explained and justified. For more in-depth literature, readers are referred to the following studies [8,13,14].

As explained before, the semantic web uses ontologies to define concepts and the relations between these concepts. Therefore, the first step in adopting this technology in any industry is developing ontologies. Therefore, the first attempts in introducing semantic web to the AEC industry have focused on developing ontologies in different industry domains to prepare foundations for future work; which would supposedly put these ontologies into use in real-life knowledge management practices. This is mainly seen in the years 2000–2012, the first decade after the semantic web invention. Multiple important ontologies have been developed in this period, a few of which are mentioned in the following.

One of the first studies in this domain was performed by Staub-French et al. [15] in 2002. They developed an ontology for relating features of building models with construction activities to facilitate cost estimation. In 2005, Lima et al. [16] proposed a high-level generic ontology, the e-COGNOS project, the first comprehensive ontology for knowledge management in the construction industry. Their effort was to develop a formal ontology for construction knowledge as a domain taxonomy for construction concepts. In construction management, semantic web repositories have been developed to support semantic knowledge management applications and facilitate interoperability over

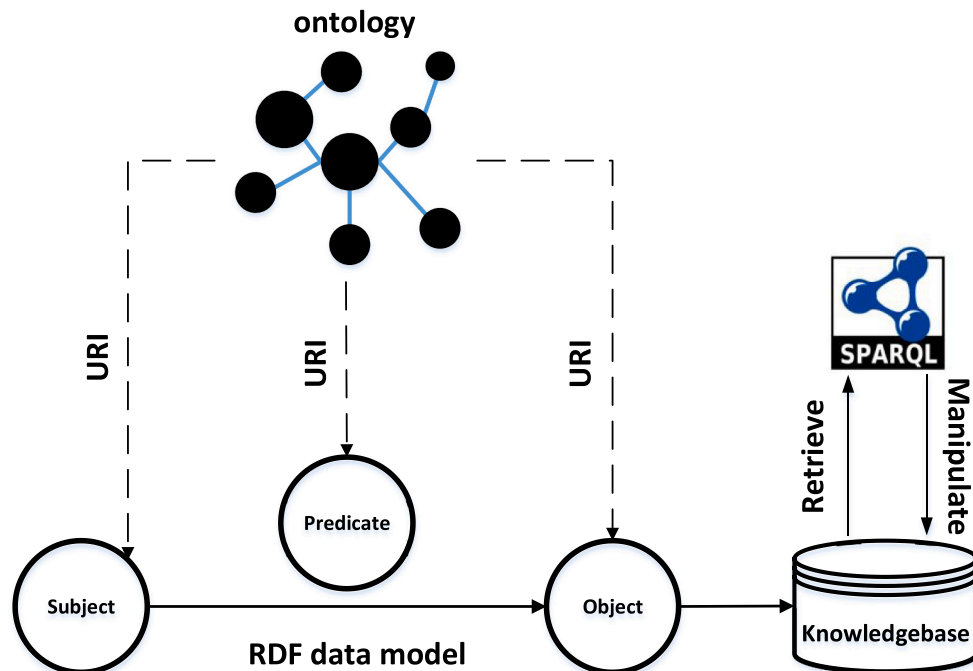


Fig. 1. Semantic Web information modeling and retrieval.

computer systems. For example, El-Diraby et al. [17] proposed an ontology, classifying construction concepts into seven major domains of Process, Product, Project, Actor, Resource, Technical Topics, and Systems. This project was used to support the e-COGNOS project by providing an ontology-based optimization of KM in e-construction. In another research, El-Diraby and Kashif [18] investigated the use of ontology architecture to model highway concepts for highway construction knowledge management. Their proposed architecture (i.e., Hi-Onto ontology) was an extension for the e-COGNOS ontology. Hi-onto linked highway geometry to highway utilities and was used as a cross-discipline exchange of Knowledge in the infrastructure domain.

In later years, studies focused on more practical applications of the semantic web in the AEC industry. For instance, in a study by Vanlande et al. [19] in 2007, a framework called C-DMF was developed based on semantic web technologies to help facilities managers organize the knowledge generated during a building's life cycle. Knowledge organization was performed in end-user contextual graphs using RDF, OWL, and the Semantic Web Rule Language (SWRL). Moving forward, more advanced ontological applications appeared. For instance, simulation modeling is one of the knowledge-intensive domains of construction engineering. Saba and Mohamed [20] highlighted the importance of reuse, composability, and interoperability of simulation models and tackled these issues using semantic web technologies. They developed a knowledge repository for storing, sharing, and managing the knowledge within construction simulation models. The repository can be employed to link the simulation models to their related information sources properly.

Studies in the first decade of the invention of the semantic web were extensively focused on developing ontologies to an extent where real-life applications were quite limited. In addition, each study proposed a different ontology for a specific domain. The proliferation of these ontologies proved that no single universally agreed-on ontology can be defined for a domain. Therefore, more recent studies are focused on integrating and combining different ontologies rather than developing an all-encompassing one. To this end, El-Gohary and El-Diraby (2009) presented an ontology integrator called Onto-Integrator to facilitate ontology interoperability within the AEC domain [21]. Their work offered a heuristic method for ontology mapping, merging concepts, relations, and axioms.

As a result, studies in the 2010s shifted from theoretical research and developing ontologies to practical research combining ontologies with other tools. The usage of the semantic web and simulation technologies in the work of Saba and Mohamed (2012) is a clear example of such a shift. Reasonably, many studies were conducted to explore the potentials of combining semantic web technologies with BIM. In the following, a number of these studies are reviewed.

2.4. Semantic web and BIM

Since the advent of BIM in the AEC industry, the application domains of this revolutionizing technology have been increasing annually. At this point, BIM is not only limited to buildings and covers other built assets such as bridges, tunnels, and roads [22,23]. Information stored in a BIM model ranges from geometrical data to building assets information and even environmental data [24]. BIM is now applied in many areas of a project with several stakeholders working with the same model. Furthermore, several BIM software applications are available in the market, each with its own unique advantages. As a result, there was a need to make collaborative design possible, and several leading software vendors initiated the concept of open BIM. The purpose was to develop a software-independent language to support a transparent, open workflow and provide enduring project data. The result was the development of the Industrial Foundation Classes or IFC format [25].

Despite the many signs of progress made in introducing, developing, and extending IFC, there were still shortcomings embedded in the information modeling layer, which was, and still is, EXPRESS-based. As

the role of BIM in linking different domain data gained prominence, the distribution of data schema and instances became a necessity. This was not easy with the major constraints of the STEP standard, such as file-based indexing of EXPRESS. The lack of a good support mechanism for data distribution called for a new approach to facilitate data exchange, mapping, reuse, and interoperability. This motivated researchers to propose the development of a semantic context for IFC [26]. In 2009, ifcOWL was introduced, which transformed the EXPRESS schema of IFC into ontologies [27].

Following the advent of ifcOWL, many studies were conducted, which can be divided into two categories: 1- how to convert the EXPRESS-based schema into ontologies [28] and 2- leveraging the ontology-based IFC to solve the AEC interoperability challenges. For instance, Barbeau et al. [29] proposed an approach to translate STEP schema and its instances to OWL. Their proposed model is called OntoSTEP. In another research, Pauwels and Terkaj (2016) looked into the existing efforts for obtaining ifcOWL ontology from EXPRESS schemas of IFC. In their paper, they presented their implementation of an EXPRESS-to-OWL converter. Similar research studies can be found in other references [30, 31]. Converting IFC to OWL enables the use of semantic web technologies and, as a result, can simplify the link to knowledge bases from other domains.

2.5. Advancing BIM interoperability

Similar to the overall adoption of the semantic web in the construction industry, more practical studies emerged after different ontologies were proposed for BIM. Since the main intention behind using semantic web technologies is to ease interoperability, these efforts mainly focused on linking data from different domains with BIM data. Some key lines of research included:

- Integration of BIM with Geographical Information System (GIS) [32]
- Using BIM for cost estimation and scheduling [33–35]
- BIM and sustainability with semantic web [36,37]

Other research topics such as BIM and manufacturing [38] or BIM-based risk management [39] also existed. Towards advancing BIM interoperability, Niknam and Karshenas [40] developed a BIM Shared Ontology (BIMSO) that enables the linkage from BIM to different domains such as scheduling [41] and cost estimation [35]. Adopting the semantic web to achieve sustainability is a relatively new topic discussed in the next section.

2.6. Semantic web & BIM – Sustainable development studies

As discussed above, the first step of applying the semantic web in any domain is to develop ontologies to model concepts. The potentials of the semantic web for sustainable development was recognized in the early 2010s in several studies. For instance, in 2013, Abanda and Keivani [8] mentioned that semantic web applications in the built environment are shifting towards sustainable approaches. In these years, numerous studies, primarily focusing on developing ontologies, explored possible ways to promote sustainability in the built environment [42,43]. Again, after developing ontologies, more practical approaches emerged. For instance, in 2018, a study proposed using BIM and semantic web to facilitate the green building evaluation processes [44]. In another study, Xiao et al. [45] proposed an information retrieval method for building energy management using semantic web and BIM technologies.

However, little work has been done to investigate the potentials of the semantic web in BIM-LCA applications. To this end, the following sections aim to review the current status of LCA in the built industry and highlight key areas where semantic web can be helpful.

2.7. LCA: tools, databases, and limitations – a case study

In this section, a brief overview of the most commonly used tools for LCA is provided. First, the concept of Environmental Product Declarations (EPDs) is introduced to explain how data is currently being gathered and distributed. Next, a case study example is presented to showcase the usability of these tools. And finally, two interviews with experts were conducted to establish the limitations and opportunities for improvement in current LCA practices.

2.7.1. LCA tools

Currently, multiple LCA tools are available developed by both the industry and the academic community. Each tool is designed based on specific countries' standards and has unique sets of features. In this section, a number of these tools, along with their features, are introduced. Here, an LCA tool refers to any environmental modeling software capable of developing and presenting LCI. The tool may conduct life cycle impact assessment (LCIA) in more complex forms, but it is not a requirement for being considered an LCA tool [46]. Generally, LCA tools use a material takeoff as input and attach it to available LCI databases. LCA tools can be classified in different ways. For instance, LCA tools can be classified into four categories of building product tools, assembly tools, or whole-building tools based on application levels. A simpler classification is based on required user skills, classifying LCA tools into LCA practitioner tools or general user tools.

In terms of application level, a product is the smallest element of analysis in "building product" tools, which are useful for evaluating and comparing building products. As an example, Building for Environmental and Economic Sustainability (BEES) is a tool that measures the environmental performance of building products [47]. Building assembly tools are more complex and can consider the combined effects of all the products within an assembly system (e.g., a wall). Athena Eco-Calculator is an example of this category. The tool is basically an excel spreadsheet with tabs for various building assemblies. In addition to providing embodied fossil energy use, the tool can also provide impact measures such as acidification and global warming potential [48]. Finally, whole-building tools bring together all the assemblies and assess the environmental impact of a building as a whole. These tools are useful for comparing different design options. The input for these tools is building geometry as well as assemblies and materials. The output is the aggregated environmental impacts of the entire building in different life cycle stages. Athena Impact Estimator is an example of this sort. The software can model over 1200 structural and envelop assembly combinations. The output can comprise the flows from and to nature, raw material flows, and emissions to air, water, and land [49].

In terms of required skills, LCA practitioner tools are mainly used to structure the analysis by linking and defining unit processes and providing analytical frameworks. As a result, the level of analysis is very detailed. Users can adjust or replace databases and conduct an LCA of individual complex products (e.g., a high-performance window assembly). In practice, these tools are used to build product/assembly LCAs to be embedded into tools such as BEES or Athena. They are not suitable to conduct a whole-building LCA, as it would be extremely demanding and practically infeasible.

On the other hand, general user tools have all the basic LCA works done in the background, with locked databases, and can only be used for building systems and products for which they have the data. As a result, users do not have to structure the analysis. However, since analytics are not designed by users, these tools require the region of the building as input to determine which databases to use. The output of these tools is often an inventory of consumed resources and emissions but can also include impact assessment [46]. In addition, users should understand the underlying working processes to be able to make sense of the outputs and their accuracy. For instance, Athena Impact Estimator uses US LCI data and is, therefore, a tool specific to North America. It is essential for users to be aware of such matters.

BIM can facilitate some of the LCA tasks. In BIM-based LCA practices, BIM models are used as data inputs for LCA tools, which solves one of the main problems of LCA practices: manual data entry. Instead of the time-consuming and error-prone process of manual data entry, all the required information for LCA will be directly provided from BIM models. These approaches often require an interoperable process allowing the exchange of models between BIM and LCA software. IFC is the primary solution in this regard, which acts as a data-sharing format between the tools; however, effective integration between LCA and BIM requires further development [50]. The tools usually conduct a quantity takeoff, extracting information directly from BIM and then combine the takeoff data with LCI data to obtain environmental metrics. Given the lack of naming standards for BIM objects and materials, it is quite possible to have inaccuracies when using the automatic takeoff features of BIM.

Studies on BIM-based LCA are still ongoing. One of the famous efforts in this regard is the LCAdesign developed in Australia by the cooperative research center for construction innovation [51]. The software takes an automatic takeoff from BIM models, input as an IFC file. It then maps the Australian LCI database to the quantities extracted from the BIM model to calculate the environmental impacts resulting from the choice of materials. Tally is another example of a BIM-based LCA tool for whole-building LCA for general users. It is an Autodesk Revit add-in that links BIM elements to the construction materials info from the Tally database and provides on-demand LCA for users to quantify the environmental impacts of building materials [52].

Table 1 provides a summary of some of the most common LCA tools available. Both classifications, as well as some important features, are provided in the table. It should be noted that each tool has a specific set of assumptions, predefined processes, and data that one should be aware of when using them. For instance, Schultz et al. compared the performance of two whole-building LCA tools, the Athena Impact Estimator (IE) and the Autodesk-Revit-based Tally plugin [53]. User input parameters, outputs, LCI databases, and user interfaces were compared in two case studies. The case studies show that there are biases in each tool resulting in discrepancies in the output results when assessing the same building. The paper concludes that users require a deeper understanding of the background processes and assumptions to take advantage of BIM-based LCA tools. This is often not the case for the average architect/engineer.

2.7.2. Environmental Product Declarations (EPDs) and Embodied Carbon in Construction Calculator (EC3)

In attempts to develop LCA tools, particular attention has been given to the embodied carbon of construction materials. Embodied carbon of a building is its carbon footprint before it becomes operational. Therefore, material embodied carbon is associated with the manufacturing of materials, including extraction and transportation to the manufacturer, the transportation of those materials to the construction site, and the construction practices used. Embodied carbon is becoming a larger portion of a building's carbon footprint as the technologies used during the operational phase of buildings are becoming more advanced. As a result, considerable efforts have been put into developing tools that facilitate the consideration of construction material embodied carbon. For instance, EC3 is one of the most advanced tools for embodied carbon calculation with a robust database of third-party verified EPDs [54].

EPD is an LCA-based method for communicating the environmental performance of a product based on the ISO 14025 standard. Building product EPDs are specifically refined by ISO 21930. Similar products with similar purposes are grouped to have a similar EPD format. These formats should follow a set of rules and requirements called Product Category Rules or PCRs. Each EPD needs to be verified by an independent third party. Cradle-to-gate LCA is usually the basis of an EPD. The verification process assesses whether an EPD is according to PCRs and standards, and evaluates the completeness and accuracy of the LCA data [55].

EC3 provides access to a robust database of EPDs from different

Table 1

Six common LCA tools and their classification.

LCA Tools	Analysis Level	User level	Important features	Link
ATHENA® Impact Estimator	Whole-Building	General	Only useful for North America - Can consider all life cycle stages - usable for both new construction and major renovations - off-shore product are not considered (they are assumed to be manufactured in North America)	http://www.athenasmi.org/our-software-data/impact-estimator/
ATHENA® EcoCalculator	Building-Assembly	General	Results are provided on a per unit area basis - obviously cannot consider operation phase - has less impact calculations as compared to IE	http://www.athenasmi.org/our-software-data/ecocalculator/
Tally	Whole-Building	General	Autodesk Revit Plugin - can consider different life cycle stages - has its own LCI - limited capacity to consider transportation	https://choosetally.com/
BEES®	Building Product	General	Its data is US average - Normalizes impact results - covers over 230 products	https://www.nist.gov/services-resources/software/bees
EC3	Material Embodied Carbon Estimator	General	Can receive BIM models as input - has considerable number of EPDs from different countries, but there are lots of countries with no data as well	https://buildingtransparency.org/ec3
SimaPro	Product Analysis	LCA Practitioners	A software for complex products with adjustable LCI databases and impact assessment methods	https://pre-sustainability.com/solutions/tools/simapro/

countries. The distribution of these EPDs over different countries can be observed in Fig. 2-a. Using EC3, designers can have an estimate of the embodied carbon impact of building materials. For instance, Fig. 2-b shows a diagram of embodied carbon for 20 MPa flowable fill concrete as well as a specific sample obtained from the United States. It should be noted that the tool is very sensitive to the location of each project. For instance, according to the tool, the average embodied carbon for a 25 MPa concrete in Asia is 355 kgCO₂e/M³, while this number is 442 kgCO₂e/M³ for the USA.

2.7.3. LCA case study

In this section, Tally will be used to conduct an LCA analysis in a case study to showcase the potentials and capabilities of the current LCA tools in the market. In addition, the analysis will act as a reference for evaluating semantic web solutions proposed later on. The building selected for this case study is a residential building located in Tehran, Iran. The building is a 12 story building, with three underground, one ground, and eight residential floors. Overall, the building has a gross floor area of 10,200 m², and a structure of reinforced concrete frames and shear walls. Fig. 3 shows the building, its information, and BIM models. The BIM models are developed using Autodesk Revit.

The owner of the project is a property developer interested in high-performance and green buildings. Given that Iran lacks LCA case studies and benchmarks for construction projects, the main purpose of LCA in this project was to establish baselines and set benchmarks for future company's projects. The project is a good candidate for this study since

it can present the abilities of current LCA tools and highlight LCA challenges in regions with little available data. The analysis provided valuable insights on the environmental impacts of the building's embodied carbon, which is often neglected in building projects in developing countries.

Once the Revit model is completed, the following inputs are given to the Tally plugin: 1- building categories to be included in the analysis, 2- expected building life-cycle, 3- transportation impacts, 4- construction impacts, and 5- operational energy impacts. Tally uses its own LCI database for material inventories and to calculate the transportation impacts. For transportation impacts, an average supply distance is given for each material. The software uses region-specific data for construction impacts and operational energy impacts; however, not every region is included. In this project, regions with similar grid mix data should be used. The inputs for the construction impacts are electricity, heating, and water. And the inputs for the operation impacts are electricity and heat, which should be calculated in an energy analysis software such as green building studio (GBS). In this project, construction impacts were not included in the analysis because of the lack of data. GBS was used as the energy analysis software since it provides easy integration with Revit software.

Table 2 shows a summary of the results of LCA analysis in Tally. The results are provided for 8 different impacts and each life cycle stage. It should be noted that the numbers are to showcase the abilities of LCA tools, and since the analysis requires region-specific data, the results are not very reliable for this project. Tally also provides a thorough report

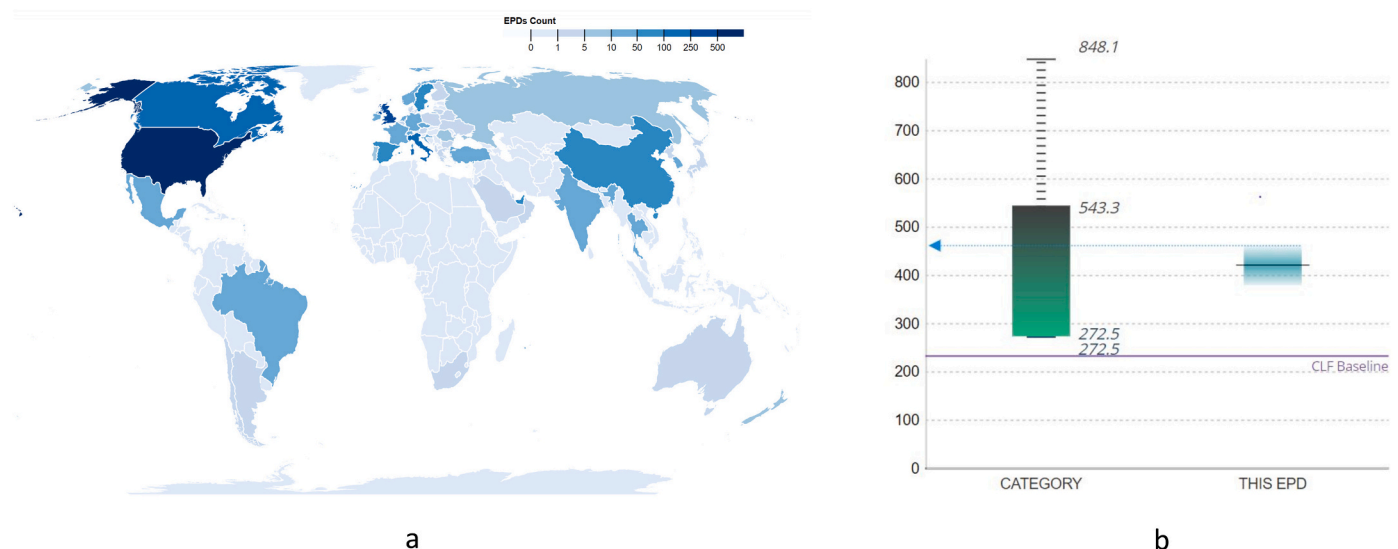
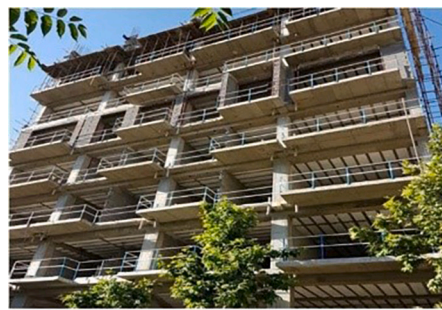
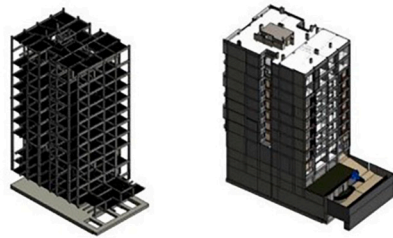


Fig. 2. a) Distribution of EPDs in the EC3 database, and b) a sample EPD obtained from [54].



Building During Construction



BIM models

Function	Residential
Floors	12 floors: 3 underground + groundfloor + 8 residential floors
Units	30
Net gross area	10,150 m ²
Location	Tehran, Iran
Completion year	2018
Owner	Nexa group
LCA Expert & Energy Modeler	Tecnosa R&D center
LCA tool	Tally
Energy modeling software	Green Building Studio
Phase in which LCA was conducted	Construction phase
Project Members involved in LCA	Owner, Architect, LCA expert/Energy Modeler

Fig. 3. Case study used in this project.

Table 2

Life Cycle Impact Assessment result summary using Tally.

Environmental Impact Totals	Product Stage [A1-A3]	Construction Stage [A4-A5]	Use Stage [B2-B4, B6]	End of Life Stage [C2-C4, D]
Acidification (kgSO ₂ eq)	14,463	382.1	66,303	1096
Eutrophication (kgNeq)	529.1	32.67	3320	178.4
Global Warming (kgCO ₂ eq)	33,71,712	72,352	2.345E+07	2,32,763
Ozone Depletion (CFC-11 eq)	0.01873	6.179E-07	1.645E-05	-2.809E-04
Smog Formation (O ₃ eq)	2,06,323	11,596	16,11,613	19,920
Primary Energy (MJ)	2.691E+07	10,32,731	3.579E+08	38,22,728
Non-renewable Energy (MJ)	2.498E+07	10,23,317	3.528E+08	37,96,475
Renewable Energy (MJ)	19,31,447	15,822	51,55,307	26,253

itemizing the impacts of different elements and materials. More will be discussed on the results in the results and discussion section.

2.7.4. LCA challenges

In this section, some of the main issues of the current LCA practices are highlighted. The purpose is to establish a baseline and explain how the semantic web can help improve the current status. To this end, in addition to the literature review and the case-study presented above, two interviews with domain experts have been conducted to enrich the section with industry expertise. The experts had 4 and 6 years of experience working as construction engineers with a research background in LCA. The discussions were open but focused on the issues facing environmental considerations in construction projects.

It is essential to be aware of the issues of current LCA tools and the impact of these issues on LCA. LCA is a very data-intensive process, and the results of the analysis are highly dependent on the availability, quality, and accuracy of input data. In general, LCA has 4 main steps 1) Goal definition and Scope, 2) Inventory Analysis, 3) Impact Assessment, and 4) Interpretation. As for the inventory, the data is region-specific as

production methods, and fuel mix differs from region to region. LCI data usually accounts for cradle-to-gate stages of materials and is represented as industry averages for whole-building LCA. Supplier/product-specific data are not usually used even if available because most LCA tools are not designed to consider them [46]. In short, using the right inventory data in an LCA is critical. However, this may turn out to be an arduous task, given that region-specific data may not be available and product/supplier-specific data may not be usable.

In the third step, the data from the inventory analysis is attributed to the impact categories to translate emissions into impacts. This process itself is data-intensive because the impact of similar emissions is not necessarily similar in different regions and environments. Once the impacts are realized, the results are provided in an informative way to interpret and evaluate whether an improvement is needed. This means that logically, LCA should result in an iterative process of correcting and improving the design. It can be imagined how difficult this may become, given that even one iteration of an LCA is quite time-consuming and resource extensive. The issue becomes even more critical once we consider that there are often little to no financial incentives for LCA [46]. Consequently, there is a lack of motivation to go through such an intensive task even for once, let alone doing it repeatedly.

According to the AIA Guide to building LCA in practice, the greatest incentive for LCA at this time is its ability to allow architects show their clients how “Green-ness” is beneficial and can increase long-term pay-backs. However, with the incompleteness and deficiencies of databases, architects must use multiple data sources and make assumptions. This makes it difficult to trust the results, as they are not reliable enough to make informed decisions with. The AIA guide further discusses that currently, there is a lack of benchmarks and guiding principles considering the unique characteristics of each building for LCA. Designers require methods that can fit well within design practices and help with specific design questions rather than current scorecard schemes that provide a list of do’s and don’ts [46]. In short, the issues with current LCA tools and practices can be either seen as process issues or data issues. A summary is provided below:

The main process issues associated with LCA applications are:

- LCA processes are not embedded in BIM tools. This means that the BIM model has to be iteratively imported into the LCA tool for any proposed design change.

- BIM and LCA software interoperability is not fully developed. For instance, the loss or change of data in transforming models into the IFC format is still an existing issue. In addition, the transformation itself may face varying technical difficulties.
- It is difficult to perform the analysis during the early stages of design. In other words, these tools only work when the design is complete. This means that once LCA is conducted, it is either too difficult, too expensive, or too time-consuming for the designers to go back and change the design. It is crucial to note that each change can impose several other changes to the design accordingly.
- Calculation and automatic inclusion of material transportation to construction sites are still unresolved [50].
- It is difficult or impossible to consider supplier/product-specific data. Also, it is almost infeasible to consider multiple suppliers for the same materials.

The main data issues associated with LCA applications are

- Lack of verified LCI data for different regions
- Lack of verified LCI data for different materials
- Insufficient data for different construction methods used in different countries. Even with similar materials, the difference in construction methods results in different environmental impacts,
- Insufficient data on material transportation and supply
- Difficult to maintain and update databases from various manufacturers and data providers

2.8. What is the gap?

Because of the issues explained above, it is quite difficult to fit an iterative LCA in the design process. Currently, LCA is conducted mostly after design completion, which means there is very little room for change and improvement. As a result, LCA often has a simple informative role rather than a decision-supporting one. To solve these issues, numerous approaches are proposed in the literature. For instance, one solution proposed by [50] is to find somehow an automatic link between BIM and accurate LCI databases and consider environmental properties in the BIM objects during the design process. This way, it would be possible to include environmental criteria in the decision-making processes in the pre-design and design phases.

Although such solutions are mostly material-oriented and do not evaluate the entire life-cycle, they propose a complementary solution to the current LCA tools. Such an approach potentially improves current LCA practices by incorporating relevant and accurate environmental information in the early design stages resulting in a more sustainable design. In addition, the improvement goes beyond just providing pre-design material data. Such data-driven approaches will also help access more accurate data for current LCA tools. Upon design completion, LCA tools can conduct a complete LCA analysis considering the entire life cycle, beyond just materials, by having reliable access to more accurate and reliable data.

So far, semantic web technology and its role and applications in the construction industry is introduced. It is highlighted that practical implementations of the semantic web are mostly performed by integrating it with BIM. One of the main topics in this regard is the semantic web's role in advancing BIM applications for sustainable development. This topic is gaining more attention in recent years; however, the potentials of semantic web technologies for promoting BIM-LCA practices is unexplored.

In this paper, we aim to examine how semantic web can be used as a solution to the data issues of LCA during the initial design stages. We will focus on the common LCA issues discussed above and investigate the extent of capabilities of the semantic web in addressing them. The study aims to set the first steps for semantic BIM-LCA integration and pave the way for future research.

The main research questions of this study are:

- Why should the semantic web be used to promote BIM-LCA applications?
- How can the semantic web be applied to a BIM-LCA analysis?
- What possible values can it provide, and how?

The first research question is partially addressed in sections 1 and 2 and will be further discussed in section 5. Sections 3 and 4 address the second research question. And sections 5 and 6 address the third research question.

3. Methodology

This section explains the process of developing a platform that can enable designers to conduct an LCA using semantic web technologies.

3.1. Determining the scope and objectives

It should be noted that this research is neither a new method to conduct LCA nor to remove all the deficiencies associated with BIM-LCA studies. The objective is solely to show how and why we can/should use semantic web technologies to facilitate the processes. LCA involves various impacts of the stages in a building life cycle, and it is an ambitious goal to cover all of its aspects. As a result, this study limits the platform to calculate material GHG emissions in the cradle-to-site stages, from material production to distribution to the project site. Extending the platform to include other aspects and stages (e.g., construction, operation and maintenance, demolition, recycling, and reuse) can be the subject of other future studies.

The platform's primary purpose is capturing the required Knowledge for conducting an LCA and accurately providing them for designers. It is expected that by doing so, 1) The required time for an LCA will be decreased, 2) LCA can be conducted in the initial stages of design, 3) LCA can be conducted with accurate information, and 4) allow simultaneous access to design information for different stakeholders. To achieve these goals, the platform has to be web-based. It also needs to store, manage, and analyze information from different sources (e.g., users, BIM, databases, online services). As such, in the proposed method, BIM and semantic web technologies are integrated into a web-based platform. The overall workflow of the platform is shown in Fig. 4. Although not addressed in this study, other knowledge sources can also be included in this platform due to the scalability of semantic web technologies, also shown in Fig. 4.

Using the proposed platform, all stakeholders can collaborate through the knowledge base. Computers can capture information from different parties, understand them, turn them into knowledge, and store them in a knowledge base accessible for other users. A knowledge base refers to a repository of readable and deducible information for machines. A knowledge base allows stakeholders to collaborate much faster and easier. For example, suppose a designer is interested to know the impacts of design. In that case, they only need to communicate with the knowledge base rather than exchanging the design information with an expert.

The process of developing the platform involves four steps of 1) Developing/reusing an ontology to use as the core schema of the knowledge base, 2) Designing a mechanism for transferring design information to the knowledge base, 3) Designing a mechanism to transfer supplier information to the knowledge base, and 4) Utilizing the knowledge base in a web-based platform to conduct LCA.

The first step aims to develop an ontology that can be used to create a knowledge base. The second step tries to establish a method to transfer design information such as the materials used and their information to the knowledge base. Since the project's scope includes the distribution of materials from suppliers, a mechanism is designed to allow users to interact and exchange supplier information with the knowledge base in the third step. Finally, step 4 explains how the knowledge base can be used to conduct LCA analytics.

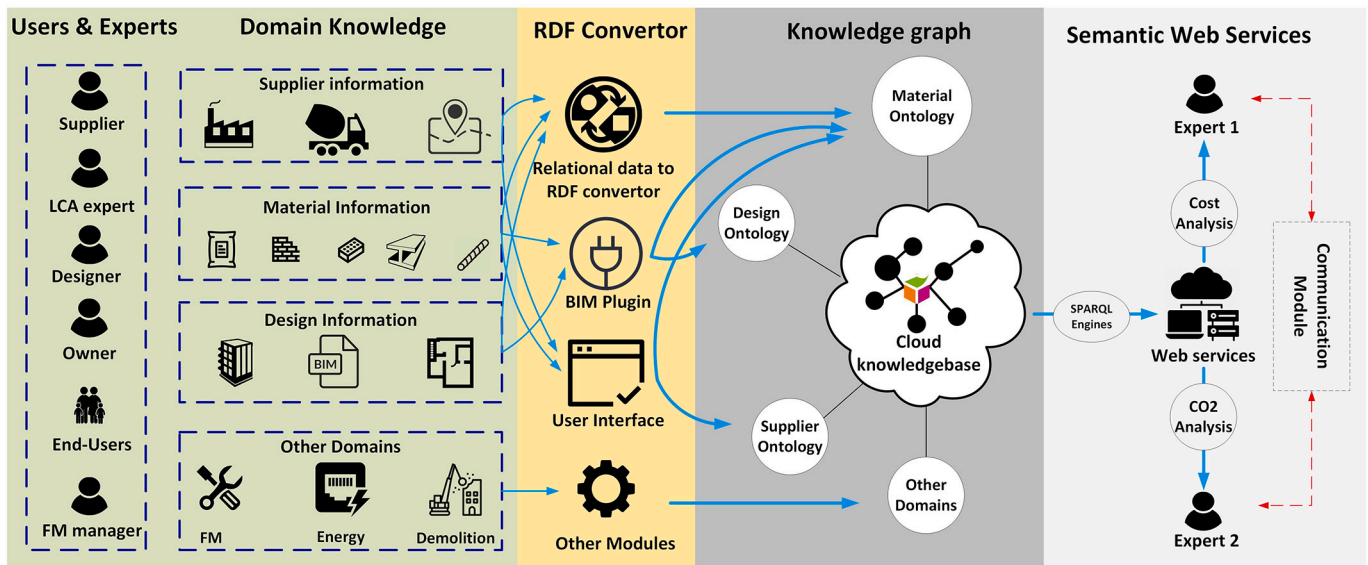


Fig. 4. The overall structure of the proposed platform.

3.2. Step 1) Developing the MATO ontology

The purpose of this step is not to develop an ontology from scratch and cover every concept. But the goal is to use an ontology to store and organize knowledge. The suggested ontology in this study is built upon a previous study that proposed a shared ontology to represent BIM data [40]. Niknam and Karshenas [40] developed a modular approach for semantic BIM modeling, allowing information integration across AEC-FM domains. This is done by designing a shared ontology called BIMSO, which consisted of concepts common to all building domains. They suggested extending the ontology to cover other applications and presented a case study for cost estimation. Therefore, BIMSO is extended using the proposed method in [40] to cover LCA's concepts. The final ontology is named MATO (material ontology) with the following considerations:

- 1- Purpose: the objective of MATO is to provide a conceptual model for cradle-to-site LCA concepts.
- 2- Scope: The scope of MATO is limited to the material GHG emissions of LCA using BIM models.
- 3- Ontology development language: RDF/OWL.
- 4- Users: experts from various building domains.

A number of concepts were added to BIMSO; the main ones are as follows:

- 1- Construction Materials: This is the main concept that should be added to the ontology since the main purpose is to calculate the GHG emissions of construction materials in the cradle-to-site stages. This class is linked to another class called "Material Property" using the "hasMaterialProperty" object property. The reverse relationship is called "hasMaterial".
- 2- Material Property: This class contains each material's information and properties in the "construction material" class. A set of data properties such as "hasDensity" and "hasThermalProperty" are defined for this class to store material properties.
- 3- Manufacturer: This class is used to store supplier information. An object property called "hasManufacturer" is defined between the "Construction Material" and "Manufacturer" classes to link each material to its suppliers. Also, a reverse relationship named "produces" is defined to link each supplier to the materials they produce. For each material produced by a manufacturer, a data property

called "hasMaterialCost" is defined for the "Manufacturer" class to store the cost of each material.

The ontology and its model in OWL are shown in Fig. 5. With MATO being developed, a framework is needed to create a semantic knowledge base and translate BIM data into a semantic web format for storage. MATO provides a schema in which BIM knowledge can be stored and used for various material analyses, particularly for LCA. Using MATO, a knowledge base can be developed in a specific and standard structure. The knowledge base stores material information such as density or embedded carbon which varies in different projects.

3.3. Step 2) Design to RDF

This step explains how designers can import their design information into the knowledge base. A key point to consider is that there are multiple design scenarios to analyze when looking for an optimal design solution. To avoid remodeling, it is suggested to create a BIM model with a low Level of Detail (LOD), such as a LOD of 200, and then create a database of design information that changes in different scenarios. This way, designers can create the BIM model's overall design and change its attributes in a database rather than frequently changing and updating the model. Furthermore, designers can find the optimum design scenario prior to modeling a BIM with high LOD and thus reduce the time and cost of design changes. As a result, designers can tell the computer the possible design options by listing design scenarios in a simple database. By capturing them in a knowledge base, the information becomes machine-readable for automated analysis.

With the design information in BIM and relational databases, a module is required to convert information into the knowledge base (RDF). To this end, this study develops a converter module, which can translate the relational information in databases and the information stored in BIM models into RDF. A BIM-based Application Programming Interface (API) is developed to access the information of BIM models. The API is developed to be flexible and allow users to decide what information about which elements should be exported. Fig. 6 shows the overall process of importing design scenarios into the knowledge base.

3.4. Step 3) Capturing supplier information

Supplier information should also be captured in the knowledge base. The followings should be considered in this process:

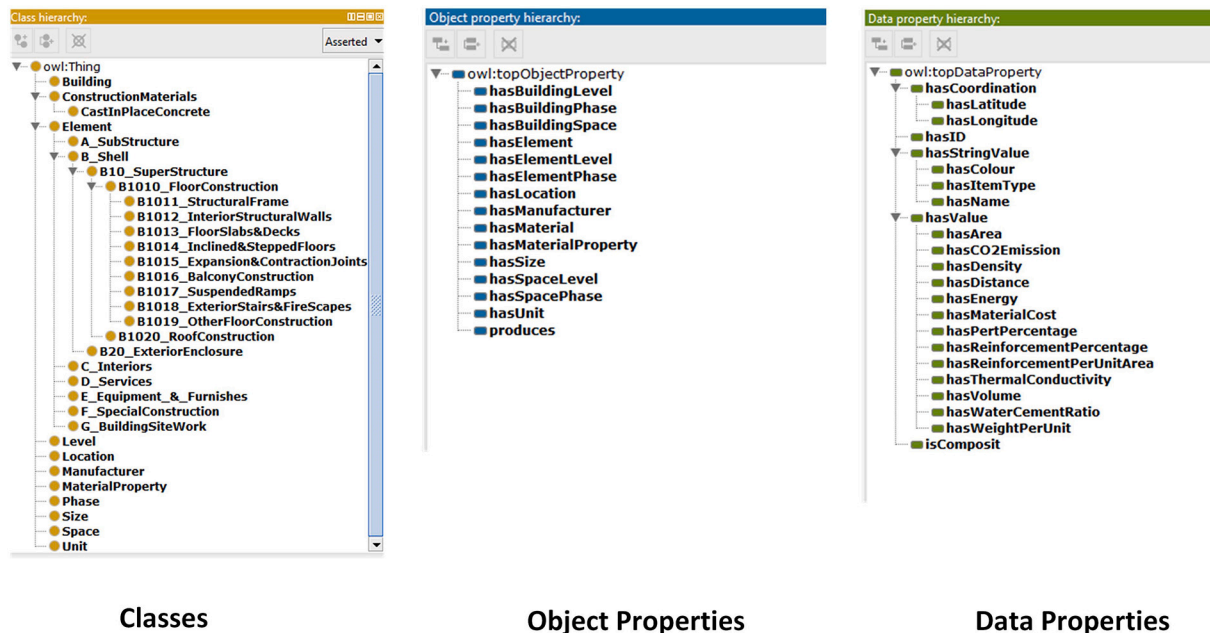


Fig. 5. MATO ontology and its implementation in Protégé.

- Users should be able to find available suppliers.
- The information of each supplier (e.g., what materials they produce, what cost, what distance from the project) should be provided for the users.
- Users should be able to query and filter suppliers.
- Users should be able to determine for which elements they intend to use the selected materials (e.g., a column or a wall).
- The information about each material (i.e., material properties) should be mapped to the supplier products.
- All the selected suppliers and their information should be stored in the knowledgebase.

Given all the above requirements, a Graphical User Interface (GUI) is needed to facilitate user interactions. Fig. 7 shows the overall process of how the proposed system works. The GUI has access to a database of suppliers that can be updated manually or automatically, using web-based APIs. Users can employ the software to access the information

in the database and query them based on their needs. The GUI has an embedded tool that can automatically tell the user the distance of each supplier to the project and the closest one. The user selects the suppliers and indicates what material they intend to purchase from each supplier. Finally, the user's preferences are exported into the knowledgebase automatically through the GUI.

3.5. Step 4) analysis

After capturing design and supplier information, the knowledge base is ready to conduct a cradle-to-site analysis to calculate materials' embedded carbon. To this end, either a single or a team of users need to access the knowledge base, retrieve the required information, and conduct the analysis. A web-based SPARQL end-point is required to enable simultaneous access for multiple users to the knowledge base. This study uses Apache Jena Fuseki Servers, which provides SPARQL 1.1 protocols for query and update. Apache Jena Fuseki allows users to

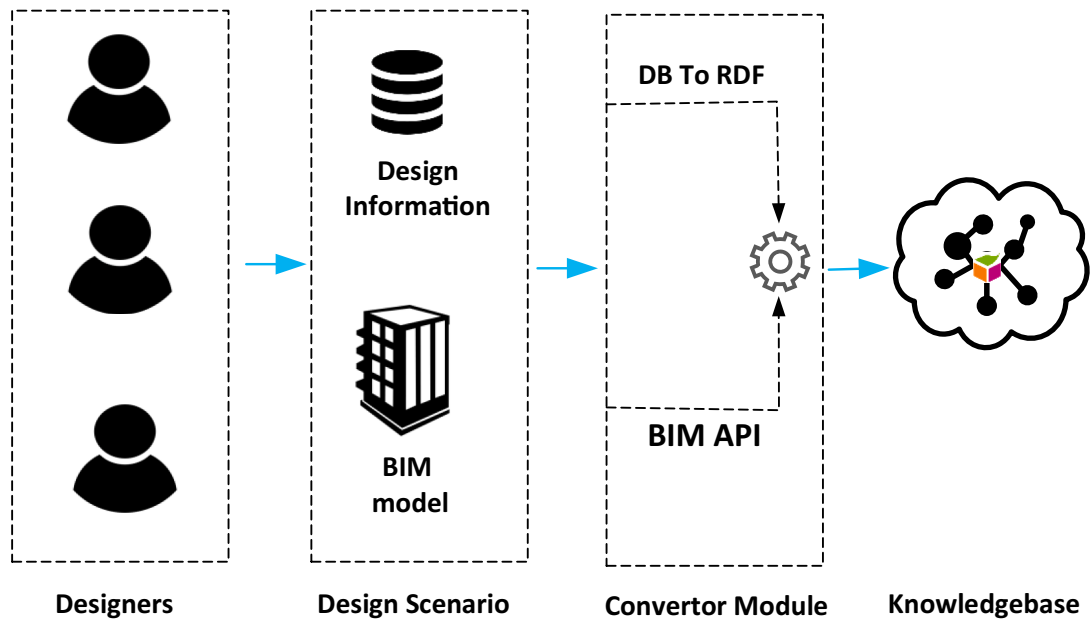


Fig. 6. The workflow for transferring design information to the knowledge base.

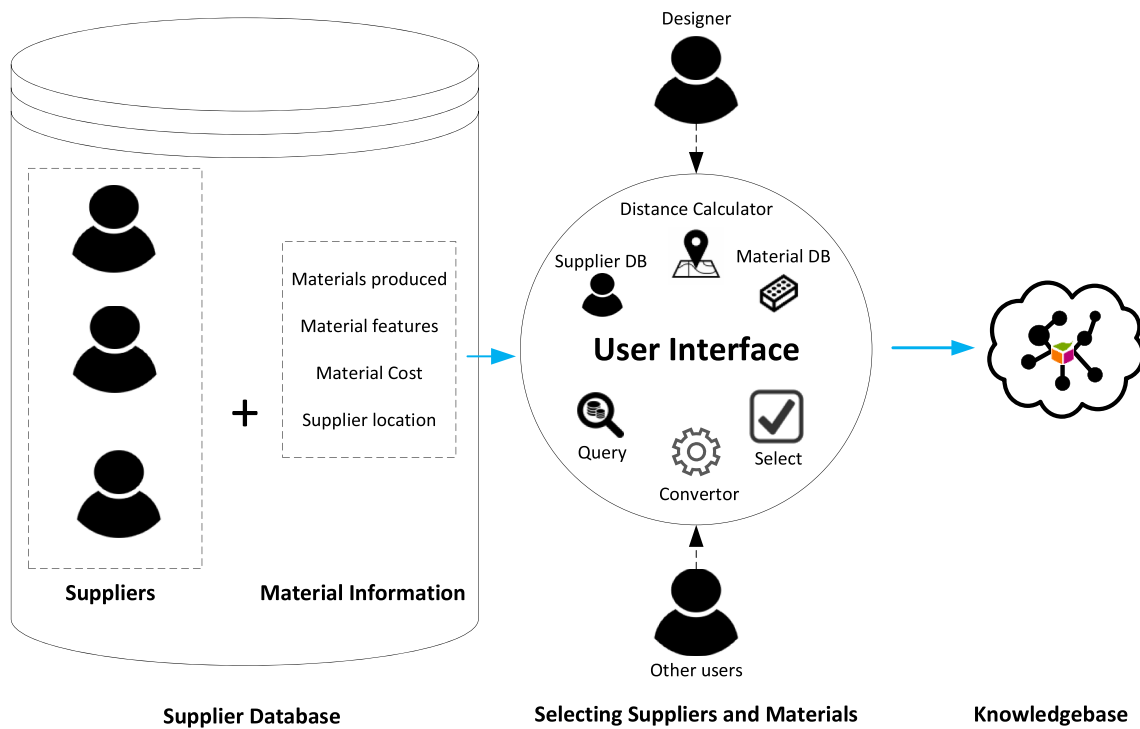


Fig. 7. The overall process of transferring supplier information to the knowledge base.

exchange their analysis results in different formats (e.g., tables, RDF, Turtle) and even share their queries directly with a link. Such a web environment allows users to interact and collaborate more efficiently. Fig. 8 shows the workflow of the proposed platform.

4. Implementation

This section aims to develop a platform based on the method described in the previous section. The purpose is to assess the usability of the proposed method and showcase how the platform can be used in real-life practices. The case study used in section 2–7–3 will be used for

this purpose. To showcase the advantages of the proposed method over current LCA practices, the platform will be used to answer design-specific issues. To this end, different design scenarios for the structural elements of the top two stories of the case study are assessed using the platform.

The scenarios are intentionally designed to be different in terms of material and supplier specifications. For instance, various concrete admixtures can be used for beams, columns, shear walls, and floors. Based on the material properties, the dimensions of structural elements change, and so does the quantity of the materials. Further, each material can be purchased from different suppliers. Obviously, analyzing the

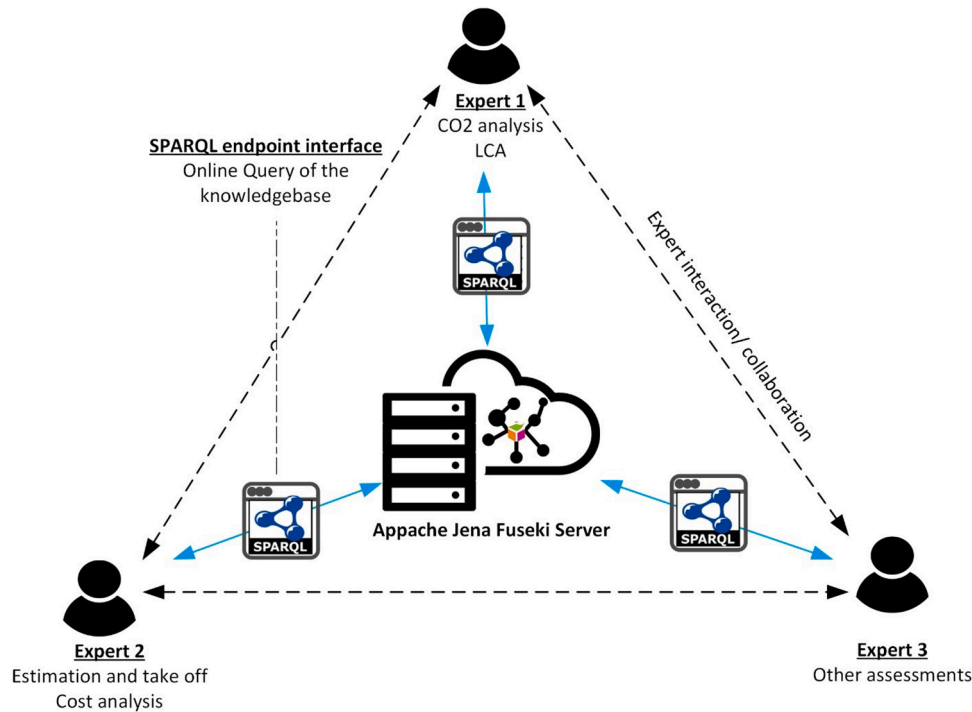


Fig. 8. The workflow of retrieving and analyzing knowledge from a knowledge base.

environmental impacts of each of these scenarios is difficult and time-consuming. The complexity of calculations and the fact that there is little motivation to conduct these sorts of analyses usually result in ignoring the environmental impacts of design scenarios or conducting them with inadequate precision.

There are 42 types of beams in the case study, 18 types of columns, and 6 types of shear walls. The platform was mainly designed to assess the concrete and reinforcement details of these elements. A design change scenario is analyzed to provide an example of how the platform works. In this scenario, the structural engineer has proposed that with a different concrete admixture, there is a design alternative for roofs in which instead of 14.33 kg/m² of rebar, 12.45 kg/m² can be used. However, the new concrete material can be purchased from a different supplier.

The following questions were designed on possible scenarios that might occur in the design and planning stages but are often answered without having an environmental perspective. The analysis here will assess the changes in the overall CO₂ emissions and costs in each scenario.

- What is the total CO₂ emission of each design option (cradle-to-site)?
- What is the material cost of each design?
- What is the effect of material supplier location on the overall CO₂ emission?

4.1. Step 1) Establishing MATO knowledge base

In the first step, we need to have some information about the environmental impact of the materials. In this case, we need to know how much CO₂ is emitted into the air as a result of producing materials. Such information should be either collected from the producers or estimated using relevant data sources. If the numbers are to be estimated, it is critical to be mindful of the region in which the material is being manufactured. For instance, the CO₂ emission of producing one cubic meter of concrete in the USA is different from Iran. In a recent study, Golzadeh [56] calculated the average embedded GHG of the production of construction materials in Iran, the results of which are used as a

reference for GHG calculations in this study. It should be noted that this study focuses on the calculation process rather than calculation results, and therefore, any other valid reference would also be reasonable to use.

4.2. Step 2) Design information to the knowledge base

To analyze different design scenarios, we need to avoid embedding all the information into the BIM model. As a result, design details that are subject to change in different design scenarios, such as the reinforcement percentage of a concrete element, are stored in a relational database. A plugin was designed to allow the designers to assign design details stored in the database from the BIM environment to facilitate the process. In summary, design information is captured in the following three ways:

- 1- Directly from BIM (e.g., Element dimensions, names, and types)
- 2- Directly from Databases (e.g., Reinforcement percentage for beam and column types)
- 3- Capturing designer preferences in the BIM environment (e.g., concrete mixture used for each element)

The process of transferring design information to the knowledge base is shown in Fig. 9. First, design details are determined in the relational database. In the BIM environment, designers use the developed plugins to determine concrete mixture details is used for each element. The plugin combines pre-determined design scenarios in the relational database (e.g., structural details for different beams) with designer choices in the BIM environment. The result will be quantity takeoffs for each type of material in each element. For documentation, the takeoff will be stored in a separate relational database. At this point, a to-RDF-Converter will convert the design info into RDF and store it in the knowledge base. This process is relatively fast and allows designers to assess different design options in a relatively short time. The RDF converter is written in a VB.Net environment using the dotNetRDF library.

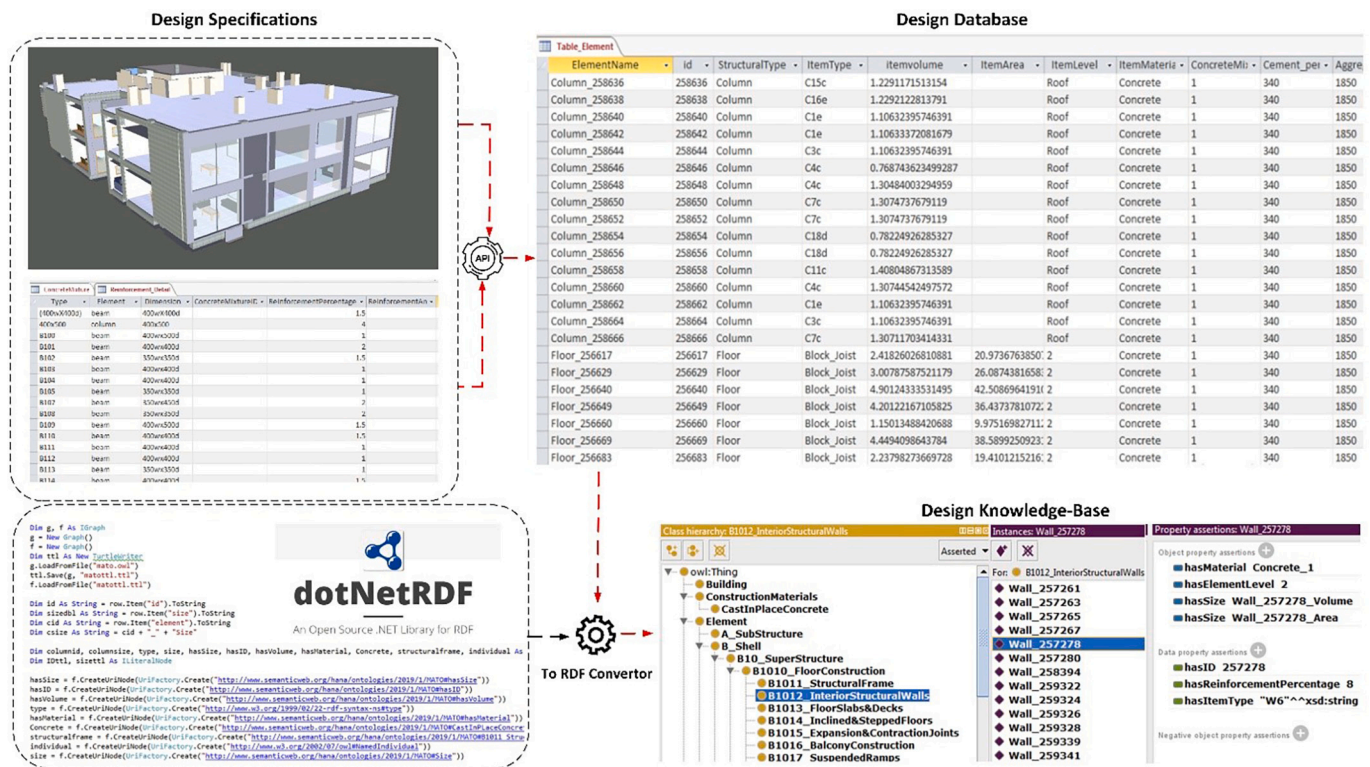


Fig. 9. Transferring design information into the knowledge base.

4.3. Step 3) Supplier information to the knowledgebase

After obtaining the design information, supplier information is required to conduct the analysis. A database of suppliers already existed for the project and is used in this project. Updating the database can be easily done manually or automatically, using web APIs. A User Interface (UI) was developed as presented in Fig. 10 to capture the supplier info from the database and present them to the designers. The UI allows users to search and query suppliers based on the materials they produce, material cost, and supplier location.

Moreover, each supplier's transportation path to the project location can be seen on a map in the UI. Designers can search for suppliers and select the ones they prefer for each material (e.g., concrete or rebars). The UI automatically converts the information into RDF and exports it into the knowledge base. Information exported into the knowledge base includes supplier location, the material supplied by the supplier, the project's location, distance, and cost.

4.4. Step 4) Analysis and results

After exporting the material, design, and supplier information, the knowledge base is ready for analysis. A web-based platform equipped with a SPARQL engine is required for the users to run their queries and collaborate. For this purpose, Apache Jena Fuseki Server is used in this project. The process is represented in [Fig. 11](#), and the results are shown in [Table 3](#). It should be noted that the knowledge base in this project was in Turtle syntax, but Fuseki server also supports N-Triples, JSON-LD, and RDF/XML.

Cradle-to-site emissions can be calculated using the total weight/volume of each material, per unit embodied carbon of materials (from [56]), project distance from suppliers, and per unit distance emissions of transporting one unit load. Fig. 11 shows a query example of how the emissions of concrete material can be calculated using the platform.

The above steps are implemented on a base design strategy. After obtaining the initial results, the analysis is conducted once again for the

alternate design option. As shown in Table 3, the two design options differ in floor structure details. Also, a different supplier was used for the second option to supply floor concrete. Overall, the results show that the alternative design option will slightly increase concrete embodied carbon, mainly because the second supplier is located further than the first supplier. However, the embodied carbon of rebars is considerably decreased. Overall, the second alternative has less environmental impacts in terms of embodied carbon from the production and transportation of materials.

5. Discussion

The proposed platform in this research has the following features:

1. It is capable of capturing knowledge in different forms in a real project. That includes: directly from users (e.g., structural details), directly captured from BIM (e.g., material takeoff), indirectly from BIM (e.g., material preferences), from relational databases (e.g., suppliers), and online services (e.g., navigation maps).
2. The platform is a web-based system, which means it has the advantages of an online system (e.g., user management and online collaboration).
3. And lastly, it is intelligent and can provide a more in-depth analysis of the information it stores. In other words, a user can run more meaningful (i.e., semantical) queries thanks to the semantic web advantages. Such an interactive relationship will ultimately result in less time and complexity in accessing information.

In the case study, users managed to access information easily, accurately, and in a relatively short time using the developed platform. Users made the following queries:

- What is the overall embodied carbon of building materials, from producing them to transporting them to the construction site?

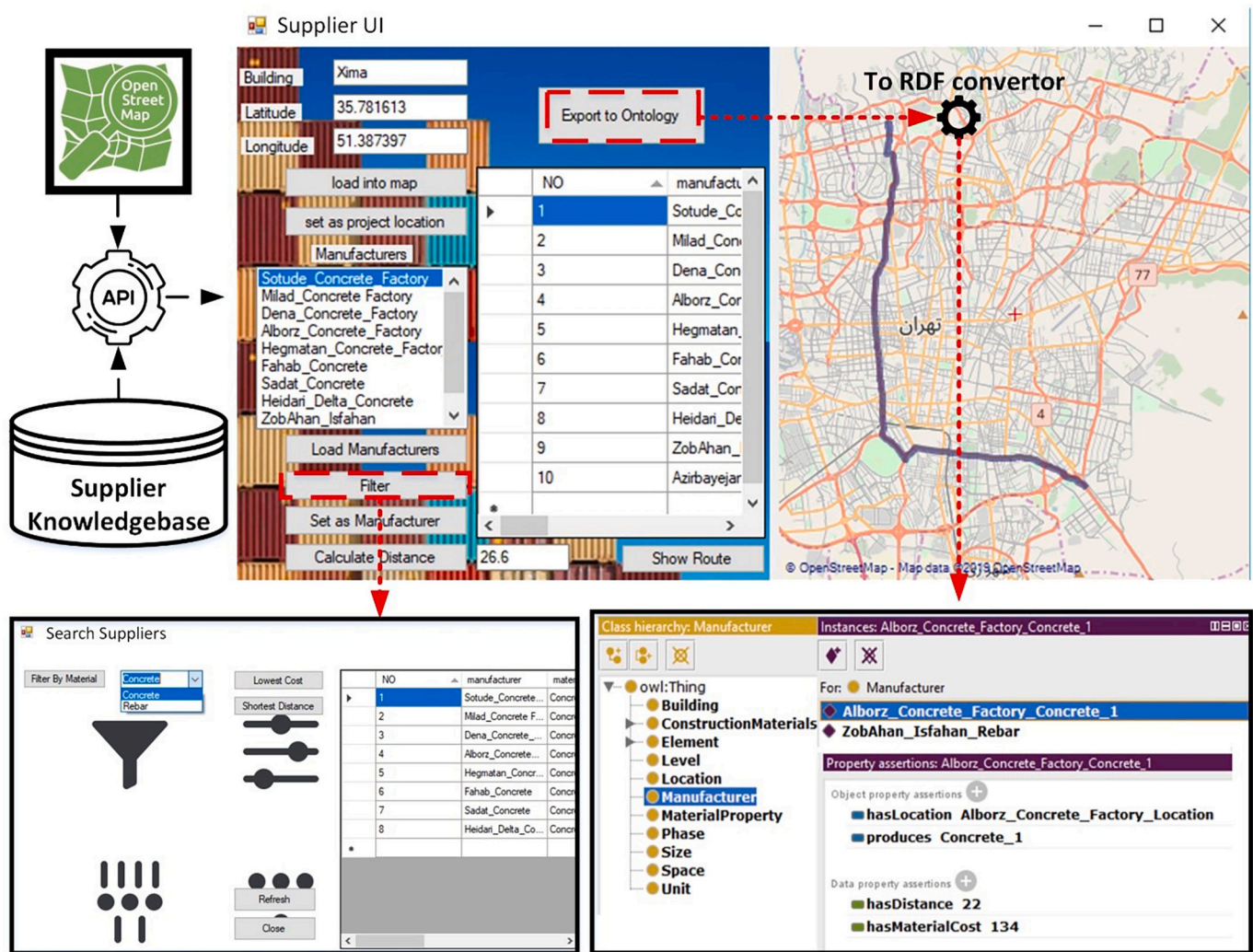


Fig. 10. The process of capturing supplier information and importing it into the knowledge base.

- What is the effect of changing the quantities of materials and suppliers?

In addition to the quick access to information, there are multiple other advantages; the most important one is the accessibility of information in the initial design stages. Thus, allowing the designers to check multiple design scenarios prior to making more detailed design decisions.

5.1. System evaluation

The system should be evaluated from three aspects. First, whether the calculations are done correctly; second, whether the results are reliable; and finally, whether the proposed system provides business value compared to other alternatives. To answer the first two questions, the results of the proposed system are compared to those generated by the Tally software, and the analysis is conducted with a simple relational database using Microsoft Access. Tally provides a detailed database of the impacts of each element in a building. The database was used to calculate the global warming potentials of the elements. The operational phase was not included in the Tally analysis as it was excluded in the semantic solution. It should also be noted that since the BIM model did not include reinforcement details for shear walls, only the concrete portion was considered for evaluation.

According to Tally, the global warming impact of the framing

elements (i.e., beams and columns) of the last two stories is 126,370 KgCO₂. The number is 18,450 KgCO₂ and 126,074 KgCO₂ for the shear walls and floors, respectively. The semantic solution yields the following numbers for the same categories: 118,167 KgCO₂ (framing), 14,779 KgCO₂ (shear walls), and 93,000 KgCO₂ (floors). The accuracy for the floor elements is 74% which is attributed to the fact that the software includes the combined effects of all the floor materials, including insulations and finishing layers which were not considered in the semantic solution. However, the accuracy of framing and wall categories is 93.5% and 80%, respectively. This accuracy can be acceptable for decision-making in the initial stages of design.

It should be noted that the two methods use different LCIs. In addition, Tally is a whole-building analysis tool which means it is capable of considering the combined effects of products in each system. The result is that the final numbers will be higher than the material-oriented analysis conducted by the semantic web solution. As for the relational database solution, the following steps were taken:

First, a set of tables were prepared to store design info, supplier info, and material info. The relationship between these tables was modeled, as shown in Fig. 12. In cases of n-to-n relationships, a third table was added for modeling the relationship. For instance, a material can have multiple manufactures, and a manufacturer might produce multiple materials.

Once the tables and the relationships between them are modeled, the calculations are performed using the Structured Query Language (SQL)

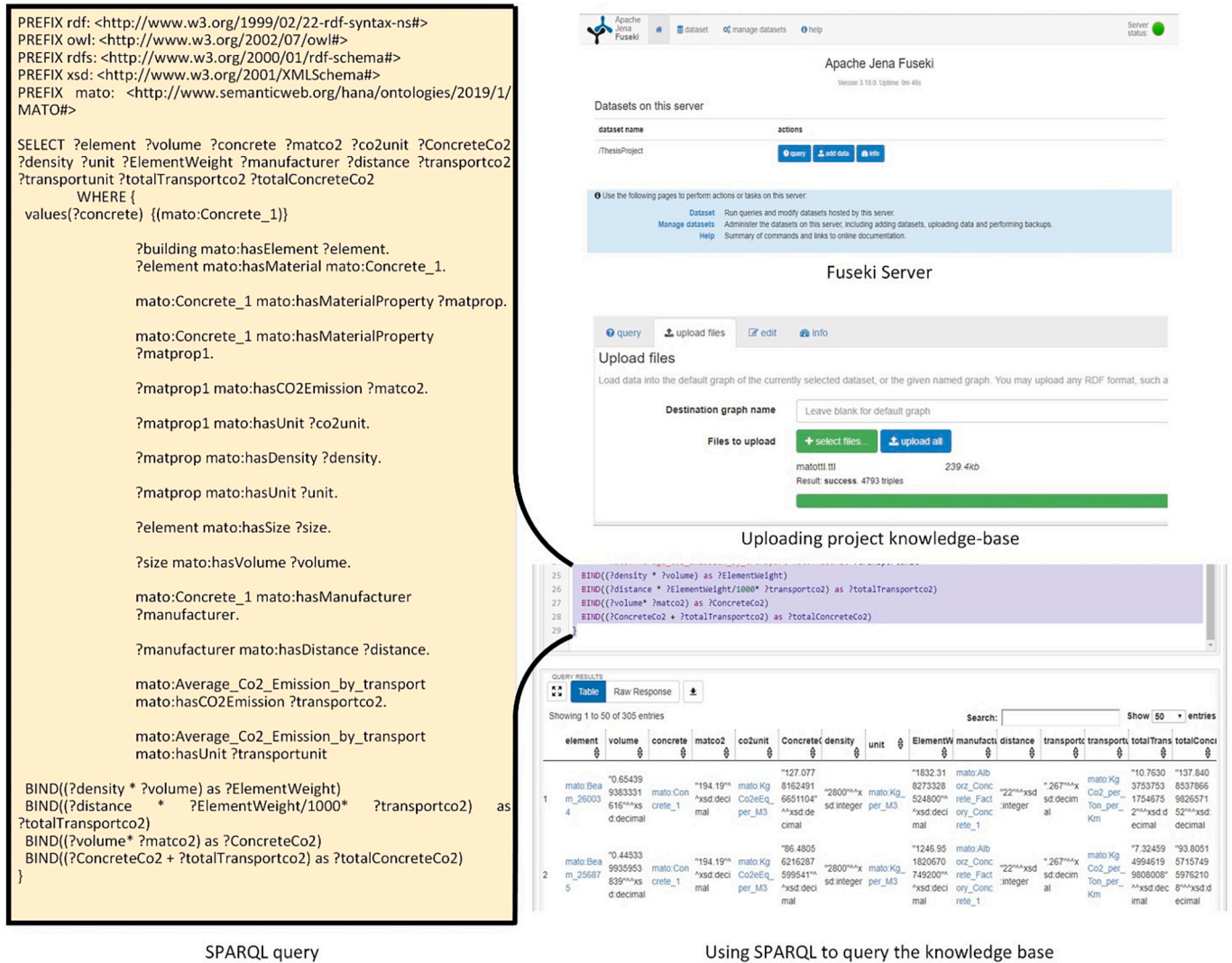


Fig. 11. Using SPARQL to calculate the CO₂ associated with producing and transporting concrete using Fuseki Server in a web-based environment.

Table 3

Final results of analyzing base design strategy and an alternative one.

Base Design CO ₂ (Kg)							
Structural Elements	Concrete			Rebar			Sum
	Weight (Kg)	CO ₂ Production	CO ₂ Transportation	Weight (Kg)	CO ₂ Production	CO ₂ Transportation	
Framing (Beam and Column)	6,54,873	52,987	3846	35,294	57,212	4122	1,18,167
Shear Wall	1,70,295	13,779	1000	38,963	63,160	4550	82,489
Floor (block joist)	5,21,697	42,211	3064	27,462	44,517	3207	92,999
Sum	13,46,865	1,08,977	7910	1,01,719	1,64,889	11,879	Sum
	1,16,887			1,76,768			
	2,93,655						
Alternative Design CO ₂ (Kg)							
Structural Elements	Concrete			Rebar			Sum
	Weight (Kg)	CO ₂ Production	CO ₂ Transportation	Weight (Kg)	CO ₂ Production	CO ₂ Transportation	
Framing (Beam and Column)	6,54,873	52,987	3846	35,294	57,212	4122	1,18,167
Shear Wall	1,70,295	13,779	1000	38,963	63,160	4550	82,489
Floor (block joist)	5,20,393	41,920	3696	23,859	38,676	2786	87,078
Sum	13,45,561	1,08,686	8542	98,116	1,59,048	11,458	Sum
	1,17,228			1,70,506			
	2,87,734						

in a process similar to the one conducted with SPARQL. Table 4 shows two of the queries performed by SQL and compares them with SPARQL. The results were the same as those obtained using SPARQL, which validates the calculations in the previous step. The results are not provided

to avoid duplication. Below, a comparison is provided to evaluate the performance of the semantic web platform:

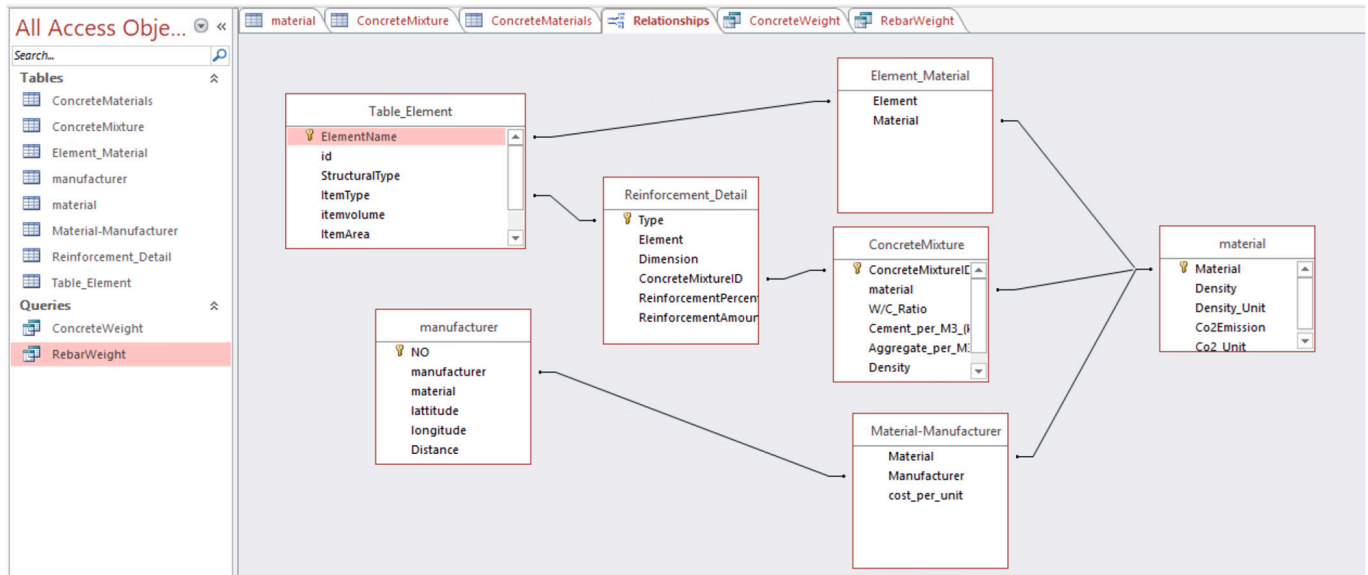


Fig. 12. Creating the data schema for analysis in Microsoft Access.

Table 4
Comparing SPARQL and SQL queries.

Query purpose	SPARQL	SQL (Access)
Calculating the weight of concrete in elements	<pre>SELECT (sum(?ElementWeight) as?TotalConcreteWeight) WHERE { ?element mato:hasMaterial mato: Concrete_1. mato:Concrete_1 mato: hasMaterialProperty?matprop. ?matprop mato:hasDensity? density. ?matprop mato:hasUnit?unit. ?element mato:hasSize?size. ?size mato:hasVolume?volume. BIND((?density *?volume) as ElementWeight) }</pre>	<pre>SELECT Sum(Table_Element. ItemVolume* ConcreteMixture.Density) AS ConcreteWeight FROM (material INNER JOIN ConcreteMixture ON material.Material = ConcreteMixture.material) INNER JOIN Table_Element ON material.Material = Table_Element. ItemMaterial;</pre>
Calculating the reinforcement weight of shear walls	<pre>SELECT (sum(?wallrebar) as? Totalwallrebar) WHERE { ?element mato:hasMaterial? material. ?element mato: hasReinforcementPercentage? ReinforcementPercentage. ?element mato:hasSize?size. ?size mato:hasVolume?Volume. ?element rdf:type mato: B1012_InteriorStructuralWalls. mato:Rebar mato: hasMaterialProperty?matprop. ?matprop mato:hasDensity? density BIND((?Volume *? ReinforcementPercentage/100* density) as?wallrebar) }</pre>	<pre>SELECT Sum(Table_Element. itemvolume* Reinforcement_Detail. ReinforcementPercentage /100*material.Density) AS Expr1 FROM Reinforcement_Detail INNER JOIN (material INNER JOIN (Table_Element INNER JOIN Element_Material ON Table_Element. ElementName = Element_Material.Element) ON material.Material = Element_Material.Material) ON Reinforcement_Detail. Type = Table_Element. ItemType WHERE (((Table_Element. ElementName) Like "*Wall*") And (Element_Material.Material = 'Rebar'));</pre>

- 1- Overall, it is much easier to use relational databases as a tool to calculate emissions. The unfamiliar structure of semantic web technologies for many users and the conversions required to use a semantic web-based knowledge base make semantic web solutions more challenging than relational databases.
- 2- Almost all the information captured in the construction industry is either stored in an unstructured way (e.g., paper documents and reports) or in the form of a relational database. As a result, using relational databases is much easier since the data is already in this form.
- 3- In terms of system complexity, the semantic solution has 11 classes (32 if subclasses are included), 15 object properties, and 23 data properties. In contrast, the relational DB solution has 7 classes (tables) with 7 links (relationships) between them.

The question that remains unanswered is why one should bother using such an unfamiliar structure of information when it even makes things more complicated? This brings us to the third aspect of the system evaluation, which is discussed in the next section.

5.2. The opportunities

First and foremost, it is essential to note that the proposed semantic-web method in this project does not aim to replace current LCA practices. Rather, it aims to provide a complementary solution for the data issues associated with current LCA tools and practices discussed in section 2–7–4. LCA is extensively data-intensive, and the availability and quality of data are crucial for reliable results. In this project, we showcased some of the capabilities of semantic-web for data management during an LCA. The applications in this project range from providing accurate LCI data, to material transportation considerations, to product/supplier-specific data consideration. In addition, it is shown that its powerful declarative query language can be embedded in the design processes making it possible to integrate different technologies and data. This means that it can be used to answer design-specific questions from the early design stages, an ability absent in current LCA practices [46].

It is not that relational databases are not able to provide such solutions, it is that they are not intentionally designed to address such meta-data-intensive tasks. A more detailed discussion is provided in section 2–2. Relational databases are the most common solution in the current LCA data management practices. Although currently, it seems to be easier to use them, there will be numerous challenges when we have

silos of data in the future. For instance, supplier and manufacturer data need to be continuously collected. In addition to the difficulties in collecting the data, it is not easy to frequently update the information and manage such large repositories.

As a result, although RDBMS seems easier and cheaper right now, it will not be so in the future, where liberal amounts of expensive workforce will be needed to manage the information. Semantic web technology is the only solution specifically designed to overcome such data challenges [12]. This is why we see emerging studies on the use of the semantic web for LCA data management. For instance, Kuczenski et al. proposed using semantic catalogs in which users can make queries to a semantic software system instead of working with a stand-alone relational database [57].

In addition, semantic services provide reasoning opportunities such as spatial reasoning and provide interpretive support. The latter can be used to link text queries to online semantic resources such as product info. Semantics are especially useful when user requirements are mostly qualitative, which is the case for many LCA challenges. Currently, users have very limited ability to discover and interpret data when dealing with qualitative requirements [57]. Semantics and data structure variations of LCA data have constrained progress in LCA fundamentally [57]. Current solutions are incapable of capturing semantics. This can be seen in the complexity difference between data schemas, where semantic-web defines over 30 classes with 15 object properties compared to the 7 tables and 7 links in the RDBMS.

As shown in Fig. 11, whenever an analyst requires to use a wide range of information with complicated relationships between them, they have to develop a data schema to store and manage the information upfront. When there are few concepts with non-complex relationships, RDBMS is the logical choice. However, this is not the case for LCA. A complete LCA covers a large number of concepts with complex relationships. Designing a schema in a relational database for such a task makes it impossible to collaborate on a full scale, where we have multiple stakeholders or different information sources.

When wide-scale collaboration is desired, intelligent systems capable of understanding data and its characteristics (i.e., meta-data) are crucial. This project was a simple example with a limited number of concepts, but in reality, many more concepts should be accounted for and included in the whole building life cycle. This is why the adoption of the semantic web is critical. Although it is not suggested to aim for a unified ontology, using ontologies to advance towards linked data approaches is suggested (e.g., the BIMSO ontology can provide a good example). Readers are referred to [12 and 33] for a more thorough analysis of the advantages of using the semantic web.

6. Summary & conclusions

This paper discusses the importance of LCA in the sustainable development of the construction industry. The discussion includes an assessment of the current LCA tools and practices with an emphasis on BIM-integrated analysis. The status quo is further demonstrated in a case study. Based on a literature review, the case study, and outreach with industry experts, the issues associated with LCA are also highlighted. It is elucidated that LCA has shown to be a difficult, expensive, and time-consuming task, with little to no financial incentives. The discussion is concluded with the need for intelligent data management solutions to address the LCA's data and meta-data intensive nature. Some of the proposed solutions in the literature were discussed, among which the integration of LCA with BIM is highlighted. However, BIM-LCA integration has also shown to have similar problems in capturing and managing information.

As a result, this research highlighted the potential benefits that the semantic web might provide for solving the data problems associated with BIM-LCA studies. It is explained that the semantic web is the only technology purposefully designed for dynamic data modeling as a metadata language dedicated to linking data of all sorts at a web-scale

[12]. LCA is an analysis that requires the integration of large complex heterogeneous databases with utmost accuracy. The semantic web is the only logical choice of technology for managing such a large-scale metadata-intensive problem. Of course, it does not mean that technologies such as relational databases cannot do the work. It is that, to do so, they have to go beyond their intended objectives, resulting in one-off, non-standard solutions [12]. This will, in turn, make integration and interoperability difficult, if not impossible, especially when we'll have silos of data in the long run.

Therefore, the study analyzes the trends in using the semantic web in the construction industry and, after justifying the existing gaps, proposes a semantic web-based platform to facilitate the data management side of LCA. The purpose is to showcase the potentials of semantic web services as complementary solutions to the current status of LCA. As a result, the platform is purposefully designed to address some of the issues identified in the literature. The focus was on the following issues: LCA in the early design stages, using LCA to address design-specific questions, and inclusion of product/supplier-specific data. This way, the paper aims to set the first steps for semantic BIM-LCA integration and pave the way for future research.

The platform was tested in a case study, and the results show that although the platform cannot perform a very accurate analysis, it has adequate accuracy to answer design-specific issues in the early stages. In addition, given its semantic capabilities, it is able to provide accurate region-specific LCI data for the well-developed LCA tools. Having access to the right data is one of the major challenges in current LCA practices, which can be resolved with semantic-web solutions. The platform's performance was also evaluated by repeating the process using relational databases. The superiority of the proposed solution in capturing semantics to relational databases is discussed in section 5–2.

The bottom line is that the integration of BIM and LCA is supposed to provide a convenient decision-making method. Such a solution should be easy to use by designers on a daily basis, or otherwise, they will not use it adequately. Currently, LCA tools are either too complex for an average designer to use or lack data accuracy. Most of them do not allow adjusting the data or do not have access to accurate regional data. The proposed project in this paper can be viewed as a starting point for the inclusion of environmental information in the early design stages. It can provide more accurate data for the LCA tools and add a data-driven decision support tool in the early design stages.

There are some limitations to the proposed semantic solutions that require additional studies. Firstly, designers should be trained to use semantic services. Further, they need to understand the environmental analysis to compare different design scenarios. Secondly, it is not possible to determine the exact point where semantic solutions will provide higher business value to the current relational database solutions as there are limited implementations of semantic solutions. Further projects are required to establish clear and understandable metrics to evaluate the business value of these solutions. Finally, the study was limited in terms of the extent of applications and domains. Further developments are required to improve and expand the proposed method to other domains and LCA stages.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to acknowledge the kind support of the Tecnosa R&D center at the University of Tehran throughout this study.

References

- [1] B. Soust-Verdaguer, C. Llatas, A. García-Martínez, Simplification in life cycle assessment of single-family houses: A review of recent developments, *Build. Environ.* 103 (2016) 215–227, <https://doi.org/10.1016/j.buildenv.2016.04.014>.
- [2] E. Krygiel, B. Nies, Green BIM: successful sustainable design with building information modeling, John Wiley & Sons, 2008, <https://doi.org/10.1111/j.1530-9290.2010.00270.x>.
- [3] R. Vanlande, C. Nicolle, C. Cruz, IFC and building life-cycle management, *Autom. Constr.* 18 (1) (2008) 70–78, <https://doi.org/10.1016/j.autcon.2008.05.001>.
- [4] J. Liebowitz (Ed.), Knowledge management handbook, CRC press, 1999, <https://doi.org/10.1201/b12285>.
- [5] I. Jurisica, J. Mylopoulos, E. Yu, Ontologies for Knowledge Management: An Information Systems Perspective, *Knowl. Inf. Syst.* 6 (2004) 380–401, <https://doi.org/10.1007/s10115-003-0135-4>.
- [6] B. Soust-Verdaguer, C. Llatas, A. García-Martínez, Critical review of bim-based LCA method to buildings, *Ener. Build.* 136 (2017) 110–120, <https://doi.org/10.1016/j.enbuild.2016.12.009>.
- [7] T. Berners-Lee, J. Hendler, O. Lassila, The Semantic Web : a new form of Web content that is meaningful to computers will unleash a revolution of new possibilities, *Sci. Am.* 284 (5) (2001) 34–43, <https://www.jstor.org/stable/26059207>.
- [8] F.H. Abanda, J.H. Tah, R. Keivani, Trends in built environment semantic Web applications: Where are we today? *Expert Syst. Appl.* 40 (14) (2013) 5563–5577, <https://doi.org/10.1016/j.eswa.2013.04.027>.
- [9] T.R. Gruber, A translation approach to portable ontology specifications, *Knowl. Acquis.* 5 (2) (1993) 199–221, <https://doi.org/10.1006/knac.1993.1008>.
- [10] M. Uschold, M. Gruninger, Ontologies: Principles, methods and applications, *Knowl. Eng. Rev.* 11 (2) (1996) 93–136, <https://doi.org/10.1017/S0269888900007797>.
- [11] M. Niknam, A semantics-based approach to construction cost estimating, PhD Dissertation, Marquette University, 2015.
- [12] J. Pollock, A semantic web business case. *Semantic Web Education and Outreach (SWEOL)*, in: W3C Semantic Web Education and Outreach interest group, 2008.
- [13] P. Pauwels, S. Zhang, Y.C. Lee, Semantic web technologies in AEC industry: A literature overview, *Autom. Constr.* 73 (2017) 145–165, <https://doi.org/10.1016/j.autcon.2016.10.003>.
- [14] J.H. Tah, H.F. Abanda, Sustainable building technology knowledge representation: Using Semantic Web techniques, *Adv. Eng. Inform.* 25 (3) (2011) 547–558, <https://doi.org/10.1016/j.aei.2011.02.006>.
- [15] S. Staub-French, M. Fischer, J. Kunz, B. Paulson, K. Ishii, An ontology for relating features of building product models with construction activities to support cost estimating, in: Center for Integrated Facility Engineering Working Paper, 2002, p. 70, [https://doi.org/10.1061/\(ASCE\)0887-3801\(2003\)17:4\(243\)](https://doi.org/10.1061/(ASCE)0887-3801(2003)17:4(243)).
- [16] C. Lima, T. El-Diraby, J. Stephens, Ontology-based optimisation of knowledge management in e-Construction, *J. Informat., Technol. Construct. (ITcon)* 10 (21) (2005) 305–327, <https://www.itcon.org/2005/21>.
- [17] T.A. El-Diraby, C. Lima, B. Feis, Domain taxonomy for construction concepts: toward a formal ontology for construction knowledge, *J. Comput. Civ. Eng.* 19 (4) (2005) 394–406, [https://doi.org/10.1061/\(ASCE\)0887-3801\(2005\)19:4\(394\)](https://doi.org/10.1061/(ASCE)0887-3801(2005)19:4(394)).
- [18] T.E. El-Diraby, K.F. Kashif, Distributed ontology architecture for knowledge management in highway construction, *J. Constr. Eng. Manag.* 131 (5) (2005) 591–603, [https://doi.org/10.1061/\(ASCE\)0733-9364\(2005\)131:5\(591\)](https://doi.org/10.1061/(ASCE)0733-9364(2005)131:5(591)).
- [19] R. Vanlande, C. Nicolle, Semantic Web technologies for facilities management, in: In 2007 2nd International Conference on Digital Information Management, 2007, pp. 327–332, <https://doi.org/10.1109/ICDIM.2007.4444244>.
- [20] F. Saba, Y. Mohamed, A Semantic Approach to Representation, Sharing, and Discovery of Construction Simulation Models. In *Construction Research Congress 2012: Construction Challenges in a Flat World*, 2012, pp. 591–601, <https://doi.org/10.1061/9780784412329.060>.
- [21] N.M. El-Gohary, T.E. El-Diraby, Merging architectural, engineering, and construction ontologies, *J. Comput. Civ. Eng.* 25 (2) (2009) 109–128, [https://doi.org/10.1061/\(ASCE\)JCP.1943-5487.0000048](https://doi.org/10.1061/(ASCE)JCP.1943-5487.0000048).
- [22] M. Marzouk, M. Hisham, S. Ismail, M. Youssef, O. Seif, On the use of building information modeling in infrastructure bridges. *CIB W78 2010 - Applications of IT in the AEC Industry*, 2010, pp. 1–10, <http://itc.scix.net/paper/w78-2010-135>.
- [23] L. Barazzetti, F. Banfi, R. Brumana, F. Roncoroni, M. Previtali, BIM from laser scans... not just for buildings: NURBS-based parametric modeling of a medieval bridge, in: XXIII ISPRS Congress, Commission V, 2016, pp. 51–56, <https://doi.org/10.5194/isprs-annals-III-5-51-2016>.
- [24] S. Azhar, Building information modeling (BIM): Trends, benefits, risks, and challenges for the AEC industry, *Leadersh. Manag. Eng.* 11 (3) (2011) 241–252, [https://doi.org/10.1061/\(ASCE\)LM.1943-5630.0000127](https://doi.org/10.1061/(ASCE)LM.1943-5630.0000127).
- [25] M. Laakso, A.O. Kiviniemi, The IFC standard: A review of history, development, and standardization, *Information Technology, ITcon* 17 (9) (2012) 134–161, <http://usir.salford.ac.uk/id/eprint/28373>.
- [26] J. Beetz, J. Van Leeuwen, B. De Vries, Towards a topological reasoning service for IFC-based building information models in a semantic web context, in: In Proceedings of Joint International Conference on Computing and Decision Making in Civil and Building Engineering, 2006, June, <http://itc.scix.net/paper/w78-2006-tf527>.
- [27] J. Beetz, J. Van Leeuwen, B. De Vries, IfcOWL: A case of transforming EXPRESS schemas into ontologies, *Ai Edam* 23 (1) (2009) 89–101, <https://doi.org/10.1017/S0890060409000122>.
- [28] P. Pauwels, W. Terkaj, EXPRESS to OWL for construction industry: Towards a recommendable and usable ifcOWL ontology, *Autom. Constr.* 63 (2016) 100–133, <https://doi.org/10.1016/j.autcon.2015.12.003>.
- [29] R. Barbau, S. Krima, S. Rachuri, A. Narayanan, X. Fiorentini, S. Fofou, R. D. Sriram, OntoSTEP: Enriching product model data using ontologies, *Comput. Aided Des.* 44 (6) (2012) 575–590, <https://doi.org/10.1016/j.cad.2012.01.008>.
- [30] Walter Terkaj, Aleksandra Soji, Ontology-based representation of IFC EXPRESS rules: An enhancement of the ifcOWL ontology, *Autom. Constr.* 57 (2015) 188–201, <https://doi.org/10.1016/j.autcon.2015.04.010>.
- [31] P. Pauwels, T. Krijnen, W. Terkaj, J. Beetz, Enhancing the ifcOWL ontology with an alternative representation for geometric data, *Autom. Constr.* 80 (2017) 77–94, <https://doi.org/10.1016/j.autcon.2017.03.001>.
- [32] E.P. Karan, J. Irizarry, Extending BIM interoperability to preconstruction operations using geospatial analyses and semantic web services, *Autom. Constr.* 53 (2015) 1–12, <https://doi.org/10.1016/j.autcon.2015.02.012>.
- [33] M. Niknam, S. Karshenas, Integrating distributed sources of information for construction cost estimating using Semantic Web and Semantic Web Service technologies, *Autom. Constr.* 57 (2015) 222–238, <https://doi.org/10.1016/j.autcon.2015.04.003>.
- [34] S.K. Lee, K.R. Kim, J.H. Yu, BIM and ontology-based approach for building cost estimation, *Autom. Constr.* 41 (2014) 96–105, <https://doi.org/10.1016/j.autcon.2013.10.020>.
- [35] M. Niknam, S. Karshenas, A semantic web service approach to construction cost estimating, *Comput. Civil Eng.* (2013) 484–491, <https://doi.org/10.1061/9780784413029.061>.
- [36] E. Petrova, P. Pauwels, K. Svidt, R.L. Jensen, Towards data-driven sustainable design: decision support based on knowledge discovery in disparate building data, *Archit. Eng. Des. Manag.* 15 (5) (2019) 334–356, <https://doi.org/10.1080/17452007.2018.1530092>.
- [37] E. Petrova, P. Pauwels, K. Svidt, R.L. Jensen, In search of sustainable design patterns: combining data mining and semantic data modelling on disparate building data, I, in: Mutis, T. Hartmann (Eds.), *Advances in Informatics and Computing in Civil and Construction, Engineering 2019*, Springer International Publishing, 2019, pp. 19–26, https://doi.org/10.1007/978-3-030-00220-6_3.
- [38] M. Niknam, F. Jalaei, S. Karshenas, Integrating BIM and product manufacturer data using the semantic web technologies, *J. Informat., Technol. Construct. (ITcon)* 24 (22) (2019) 424–439, <https://www.itcon.org/2019/22>.
- [39] L.Y. Ding, B.T. Zhong, S. Wu, H.B. Luo, Construction risk knowledge management in BIM using ontology and semantic web technology, *Saf. Sci.* 87 (2016) 202–213, <https://doi.org/10.1016/j.ssci.2016.04.008>.
- [40] M. Niknam, S. Karshenas, A shared ontology approach to semantic representation of BIM data, *Autom. Constr.* 80 (2017) 22–36, <https://doi.org/10.1016/j.autcon.2017.03.013>.
- [41] M. Niknam, S. Karshenas, Integrating BIM and Project Schedule Information Using Semantic Web Technology, *Construct. Res. Congr.* 2016 (2016) 689–697, <https://doi.org/10.1061/9780784479827.070>.
- [42] J.H. Tah, H.F. Abanda, Sustainable building technology knowledge representation: Using Semantic Web techniques, *Adv. Eng. Inform.* 25 (3) (2011) 547–558, <https://doi.org/10.1016/j.aei.2011.02.006>.
- [43] F.H. Abanda, J.H. Tah, D. Duce, PV-TONS: A photovoltaic technology ontology system for the design of PV-systems, *Eng. Appl. Artif. Intell.* 26 (4) (2013) 1399–1412, <https://doi.org/10.1016/j.engappai.2012.10.010>.
- [44] S. Jiang, N. Wang, J. Wu, Combining BIM and ontology to facilitate intelligent green building evaluation, *J. Comput. Civ. Eng.* 32 (5) (2018), [https://doi.org/10.1061/\(ASCE\)JCP.1943-5487.0000786](https://doi.org/10.1061/(ASCE)JCP.1943-5487.0000786), 04018039.
- [45] Y. Xiao, Z. Hu, J. Lin, Ontology-Based Semantic Retrieval Method of Energy Consumption Management. *Advances in Informatics and Computing in Civil and Construction Engineering*, Springer, 2019, pp. 231–238, https://doi.org/10.1007/978-3-030-00220-6_28.
- [46] C. Bayer, M. Gamble, R. Gentry, S. Joshi, AIA Guide to building life cycle assessment in practice, The American Institute of Architects, Washington DC, 2010. Available: <https://www.aia.org/resources/7961-building-life-cycle-assessment-in-practice>.
- [47] BEES LCA software, Available, <https://www.nist.gov/services-resources/software/bees>. Accessed Date: 29 December 2020.
- [48] Athena Eco-Calculator Software, Available, <http://www.athenasmi.org/our-software-data/ecocalculator>. Accessed Date: 29 December 2020.
- [49] Athena Impact Estimator Software, Available, <http://www.athenasmi.org/our-software-data/impact-estimator>. Accessed Date: 29 December 2020.
- [50] L.Á. Antón, J. Díaz, Integration of life cycle assessment in a BIM environment, *Procedia Eng.* 85 (2014) 26–32, <https://doi.org/10.1016/j.proeng.2014.10.525>.
- [51] S. Tucker, M. Ambrose, D. Johnston, P. Newton, S. Seo, D. Jones, LCA Design: An integrated approach to automatic eco-efficiency assessment of commercial buildings, *CIB W078 conference*, 2003 <https://eprints.qut.edu.au/27429>, Accessed Date: 22 July 2021.
- [52] L.C.A. Tally, Revit Plugin, Available: <https://choosetally.com>.
- [53] J. Schultz, K. Ku, M. Gindlesparger, J. Doerfler, A benchmark study of BIM-based whole-building life-cycle assessment tools and processes, *Int. J. Sustain. Build. Technol. Urban Dev.* 7 (3–4) (2016) 219–229, <https://doi.org/10.1080/2093761X.2017.1302839>.
- [54] Embodied Carbon in Construction Calculator (EC3) Tool, Available, <https://buildingtransparency.org/ec3>. Accessed Date: 4 January 2020.

- [55] EPDs – Type III labels to ISO 14025, Available, <https://www.greenspec.co.uk/epds-type-iii-labels-to-iso-14025>. Accessed Date: 4 January 2021.
- [56] P. Golzadeh, A study on the impacts of construction materials of concrete buildings based on sustainable development indexes, MSc Thesis, University of Tehran, 2016. <https://ut.ac.ir/fa/thesis/15957>, Accessed Date: 22 July 2021..
- [57] B. Kuczenski, C.B. Davis, B. Rivela, K. Janowicz, Semantic catalogs for life cycle assessment data, J. Clean. Prod. 137 (2016) 1109–1117, <https://doi.org/10.1016/j.jclepro.2016.07.216>.